

CSCI596: Scientific Computing & Visualization

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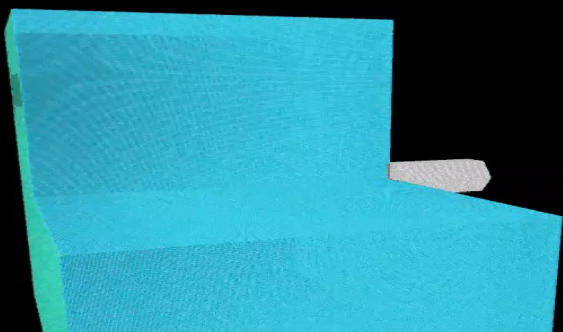
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- **Computational science:** An area of scientific investigation, where computers play a central role —do science using computer (*e.g.*, computational chemistry).
- **Scientific computing:** An area in computer science (CS) to support computational sciences by innovative use of computer systems—parallel computing, scientific visualization, *etc.*



Hierarchical Computing Architecture

4.9 trillion-atom molecular dynamics
40 trillion-d.o.f. quantum mechanics

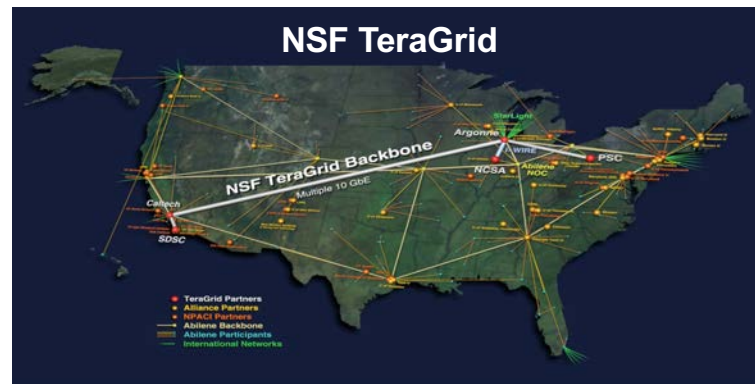


Exaflop/s computers

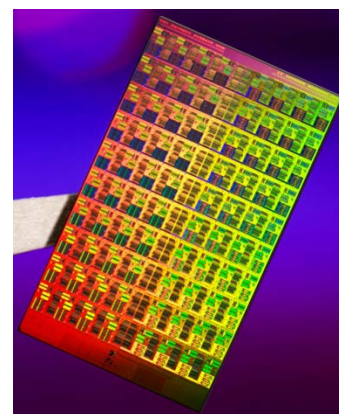


Exaflop/s = 10^{18} floating-point operations per second

Global Grid of supercomputers



Many-core CPU computing



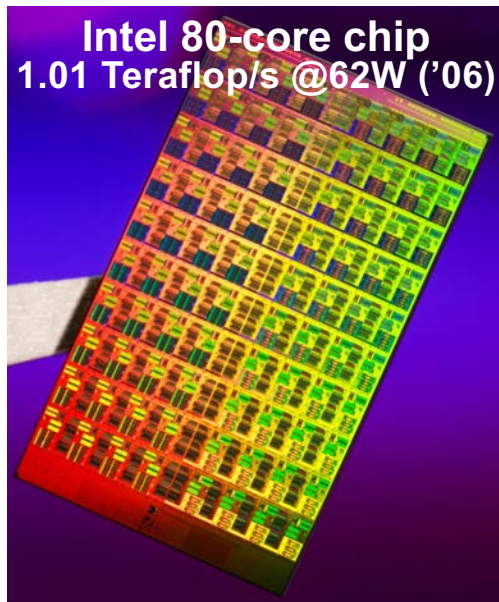
Intel 80-core chip
1Tflop/s@62W ('06)

10^2 -cluster
Grid

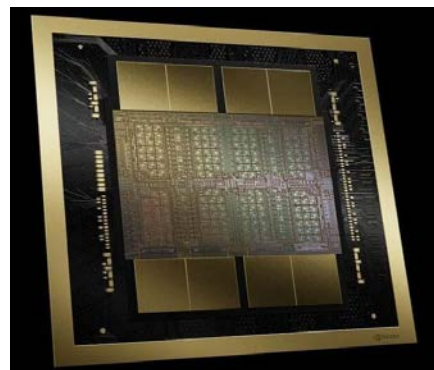
10^5 -node
cluster

10^3 -core
node

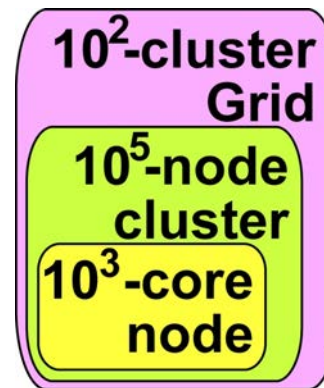
Many-core CPU/GPU Computing



ASCI Red ('97)
Teraflop/s @1MW

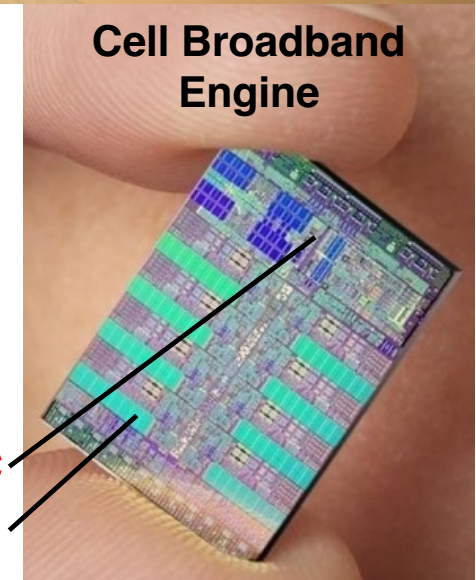


NVIDIA B200 GPU ('24)
37,888 CUDA cores
592 Tensor cores
20,000 Teraflop/s @1KW



64bit PowerPC

8 synergistic processing elements



World's Biggest Company

Top 10 Companies by Market Capitalization

- **Nvidia:** ~\$4.8 trillion to \$5.0 trillion (U.S. - Semiconductors/AI)
- **Apple:** ~\$4.3 trillion to \$4.9 trillion (U.S. - Consumer Electronics)
- **Alphabet (Google):** ~\$3.8 trillion to \$3.9 trillion (U.S. - Technology/Services)
- **Microsoft:** ~\$2.8 trillion (U.S. - Software/Cloud)
- **Amazon:** ~\$2.5 trillion (U.S. - E-commerce/Cloud)
- **TSMC:** ~\$2.1 trillion (Taiwan - Semiconductors)
- **Broadcom:** ~\$1.8 trillion (U.S. - Semiconductors)
- **Saudi Aramco:** ~\$1.7 trillion (Saudi Arabia - Energy/Oil)
- **SpaceX:** ~\$1.5 trillion (U.S. - Aerospace)
- **Meta Platforms:** ~\$1.5 trillion (U.S. - Technology/Social Media)

July 24, 2026



Jensen Huang



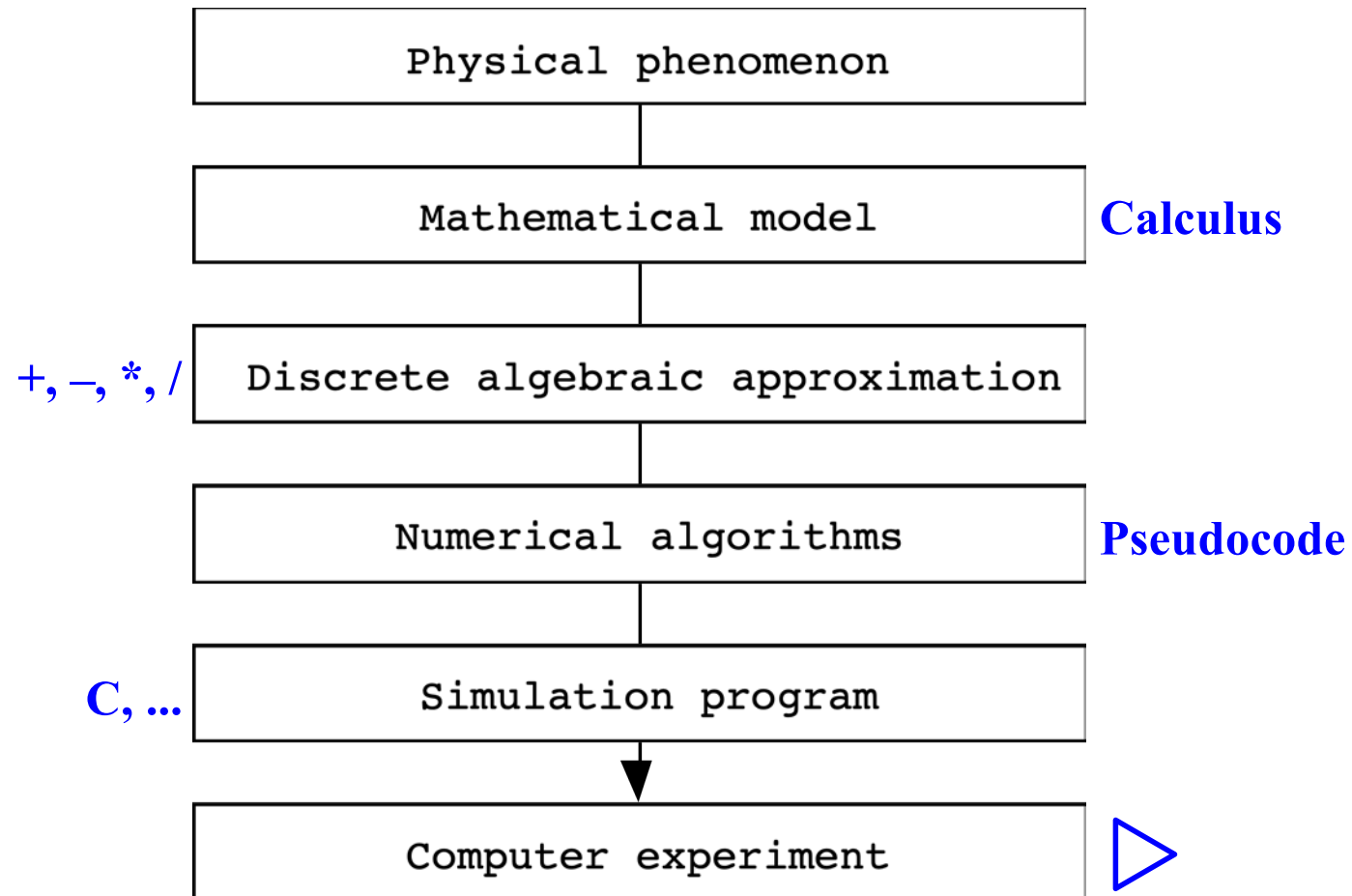
Mar. 26, 2026, Taipei

Computer Simulation

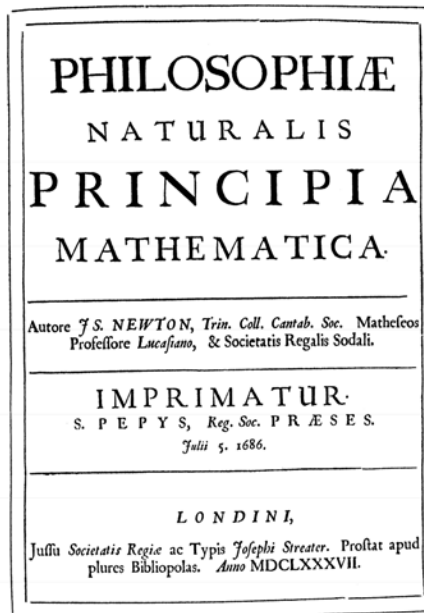
Run real applications on advanced computing architectures



Computer Experiment



Mathematical Model



TITLE PAGE OF THE FIRST EDITION OF THE PRINCIPIA
(See Appendix, Note 1, para 6-7)

AXIOMS, OR LAWS OF MOTION¹

LAW I

Every body continues in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it.

PROJECTILES continue in their motions, so far as they are not retarded by the resistance of the air, or impelled downwards by the force of gravity. A top, whose parts by their cohesion are continually drawn aside from rectilinear motions, does not cease its rotation, otherwise than as it is retarded by the air. The greater bodies of the planets and comets, meeting with less resistance in freer spaces, preserve their motions both progressive and circular for a much longer time.

LAW II²

The change of motion is proportional to the motive force impressed; and is made in the direction of the right line in which that force is impressed.

If any force generates a motion, a double force will generate double the motion, a triple force triple the motion, whether that force be impressed altogether and at once, or gradually and successively. And this motion (being always directed the same way with the generating force), if the body moved before, is added to or subtracted from the former motion, according as they directly conspire with or are directly contrary to each other; or obliquely joined, when they are oblique, so as to produce a new motion compounded from the determination of both.

LAW III

To every action there is always opposed an equal reaction: or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

Whatever draws or presses another is as much drawn or pressed by that other. If you press a stone with your finger, the finger is also pressed by the

¹ Appendix, Note 14. ² Appendix, Note 15.
[13]

LAW I

Every body continues in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it.

LAW II²

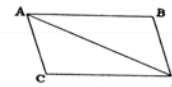
The change of motion is proportional to the motive force impressed; and is made in the direction of the right line in which that force is impressed.

stone. If a horse draws a stone tied to a rope, the horse (if I may so say) will be equally drawn back towards the stone; for the distended rope, by the same endeavor to relax or unbend itself, will draw the horse as much towards the stone as it does the stone towards the horse, and will obstruct the progress of the one as much as it advances that of the other. If a body impinge upon another, and by its force change the motion of the other, that body also (because of the equality of the mutual pressure) will undergo an equal change, in its own motion, towards the contrary part. The changes made by these actions are equal, not in the velocities but in the motions of bodies; that is to say, if the bodies are not hindered by any other impediments. For, because the motions are equally changed, the changes of the velocities made towards contrary parts are inversely proportional to the bodies. This law takes place also in attractions, as will be proved in the next Scholium.

COROLLARY I

A body, acted on by two forces simultaneously, will describe the diagonal of a parallelogram in the same time as it would describe the sides by those forces separately.

If a body in a given time, by the force M impressed apart in the place A, should with an uniform motion be carried from A to B, and by the force N impressed apart in the same place, should be carried from A to C, let the parallelogram ABCD be completed, and, by both forces acting together, it will in the same time be carried in the diagonal from A to D. For since the force N acts in the direction of the line AC, parallel to BD, this force (by the second Law) will not at all alter the velocity generated by the other force M, by which the body is carried towards the line BD. The body therefore will arrive at the line BD in the same time, whether the force N be impressed or not; and therefore at the end of that time it will be found somewhere in the line BD. By the same argument, at the end of the same time it will be found somewhere in the line CD. Therefore it will be found in the point D, where both lines meet. But it will move in a right line from A to D, by Law 1.



New Kind of Science

Some 400 years later, Newton, in his efforts to understand the natural laws of the rate of change in motion, used algebra to underpin another new branch of mathematics: calculus (a branch for which von Leibniz is simultaneously and independently credited). Calculus spurred scientists “to go off looking for other laws of nature that could explain natural phenomenon in terms of rates of change and found them by the bucketful – heat, sound, light, fluid dynamics, electricity and magnetism” [2].

A scientific revolution is just beginning.

However, what this report uncovers, for the first time, is a fundamentally important shift from *computers* supporting scientists to ‘do’ traditional science to computer science becoming embedded into the very fabric of science and how science is done, creating what I am prepared to go so far as to call ‘new kinds’ of science¹.

Stephen Emmott

<https://www.microsoft.com/en-us/research/publication/towards-2020-science-2>

<http://www.nature.com/nature/journal/v440/n7083>

Molecular Dynamics Simulation

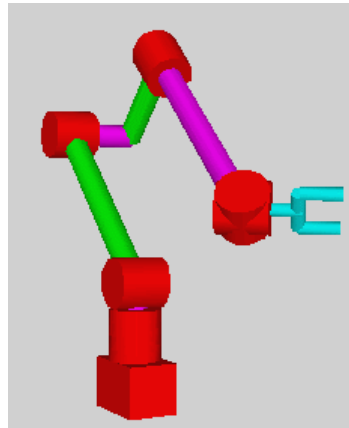
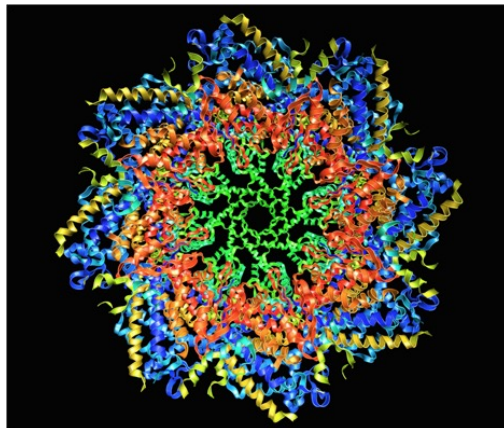
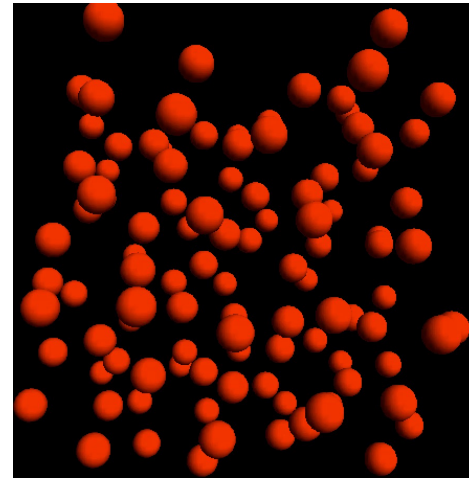
- Newton's equation of motion

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = - \frac{\partial V(\mathbf{r}^N)}{\partial \mathbf{r}_i} \quad (i = 1, \dots, N)$$

- Many-body interatomic potential

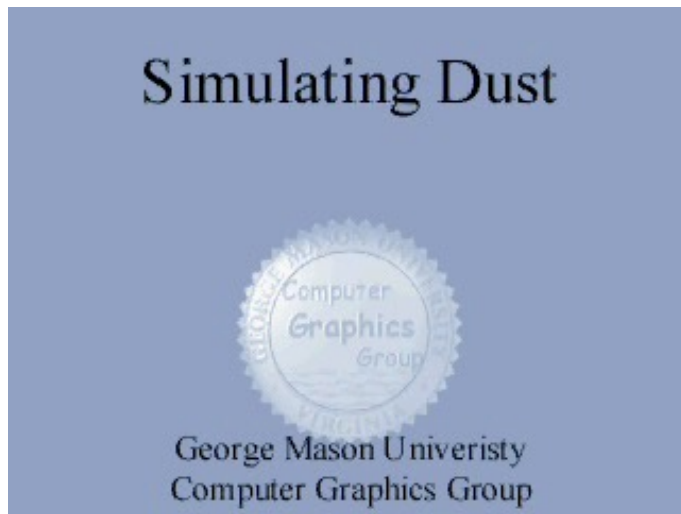
$$V = \sum_{i < j} u_{ij}(|\mathbf{r}_{ij}|) + \sum_{i, j < k} v_{jik}(\mathbf{r}_{ij}, \mathbf{r}_{ik})$$

- Application: drug design, robotics, entertainment, *etc.*



Molecular Dynamics in Graphics

Particle modeling of dust & smoke



Jim Chen (George Mason)

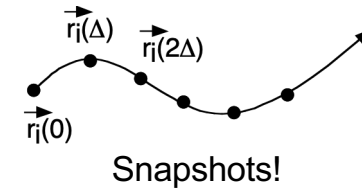
Ron Fedkiw (Stanford)



MD Algorithm

Time discretization: differential → algebraic equation

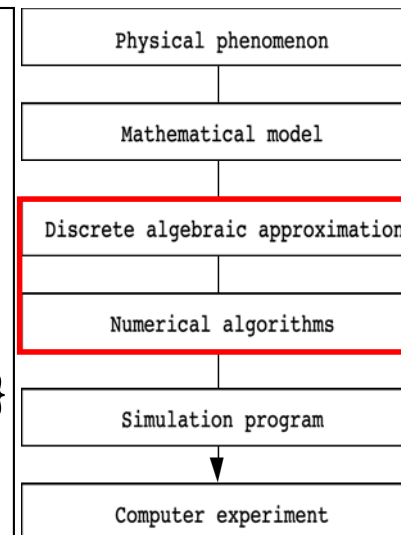
$$\begin{cases} \vec{r}_i(t + \Delta) = \vec{r}_i(t) + \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 \\ \vec{v}_i(t + \Delta) = \vec{v}_i(t) + \frac{\vec{a}_i(t) + \vec{a}_i(t + \Delta)}{2}\Delta \end{cases} \quad \vec{a}_i = -\frac{1}{m} \frac{\partial V}{\partial \vec{r}_i}$$



Time stepping: Velocity Verlet algorithm

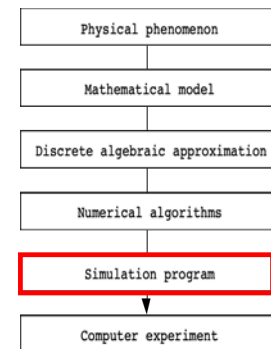
Given $(\vec{r}_i(t), \vec{v}_i(t))$

1. Compute $\vec{a}_i(t)$ as a function of $\{\vec{r}_i(t)\}$
2. $\vec{v}_i\left(t + \frac{\Delta}{2}\right) \leftarrow \vec{v}_i(t) + \frac{\Delta}{2}\vec{a}_i(t)$
3. $\vec{r}_i(t + \Delta) \leftarrow \vec{r}_i(t) + \vec{v}_i\left(t + \frac{\Delta}{2}\right)\Delta$
4. Compute $\vec{a}_i(t + \Delta)$ as a function of $\{\vec{r}_i(t + \Delta)\}$
5. $\vec{v}_i(t + \Delta) \leftarrow \vec{v}_i\left(t + \frac{\Delta}{2}\right) + \frac{\Delta}{2}\vec{a}_i(t + \Delta)$



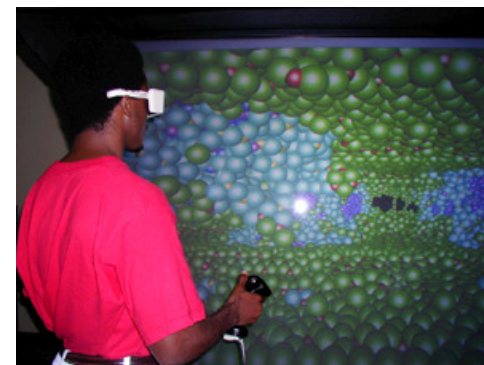
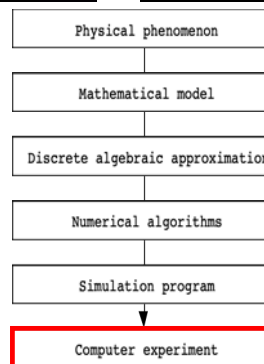
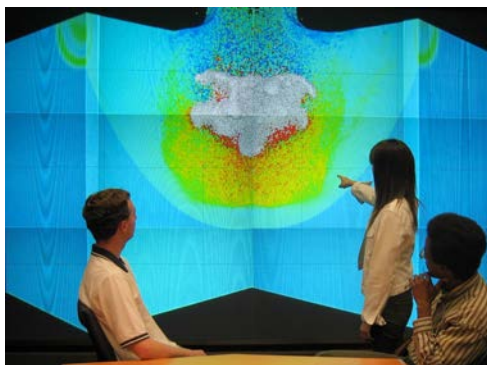
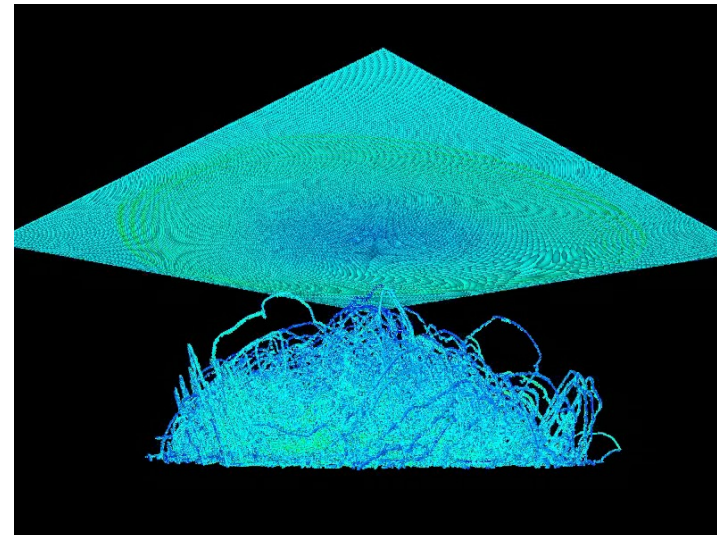
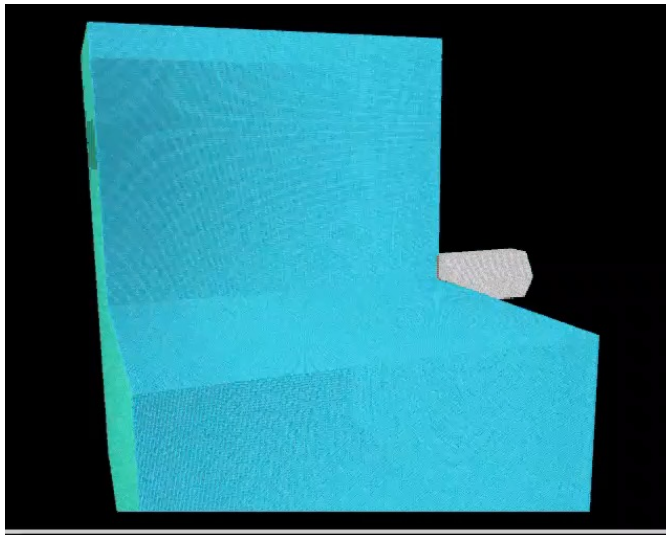
MD Program

```
for (n=0; n<nAtom; n++)  
    for (k=0; k<3; k++)  
        rv[n][k] = rv[n][k] + DeltaT/2*ra[n][k];  
  
for (n=0; n<nAtom; n++)  
    for (k=0; k<3; k++)  
        r[n][k] = r[n][k] + DeltaT*rv[n][k];  
  
ComputeAccel();    // r[][] → ra[][]  
  
for (n=0; n<nAtom; n++)  
    for (k=0; k<3; k++)  
        rv[n][k] = rv[n][k] + DeltaT/2*ra[n][k];
```



Computer Experiment

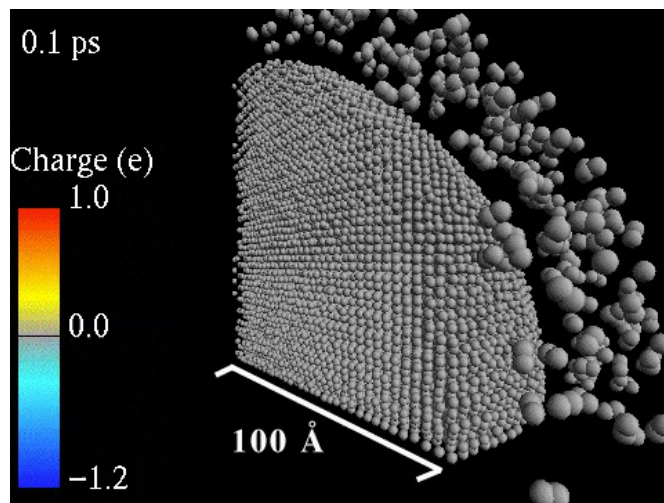
200-500 million atom molecular dynamics simulation of hypervelocity (15 km/s) impact on AlN & Al₂O₃ plate



Type of Mathematical Models

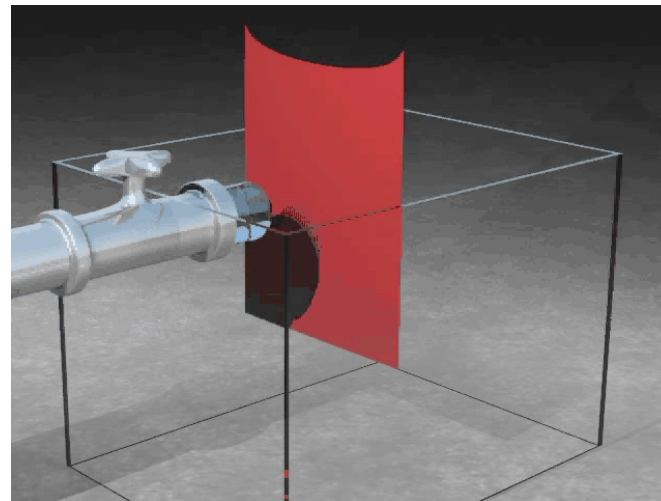
	Particle model (ordinary differential equations)	Continuum model (partial differential equations)
Deterministic	molecular dynamics	computational fluid dynamics, continuum mechanics
Stochastic	Monte Carlo particle simulation	quantum Monte Carlo

Particle model of oxidation



Tim Campbell
USC-CACS

Continuum model of water flow



Ron Fedkiw (Stanford)
graphics.stanford.edu/~fedkiw

Continuum Model: Quantum Mechanics

Challenge: Complexity of quantum N -body problem

Density functional theory (DFT)

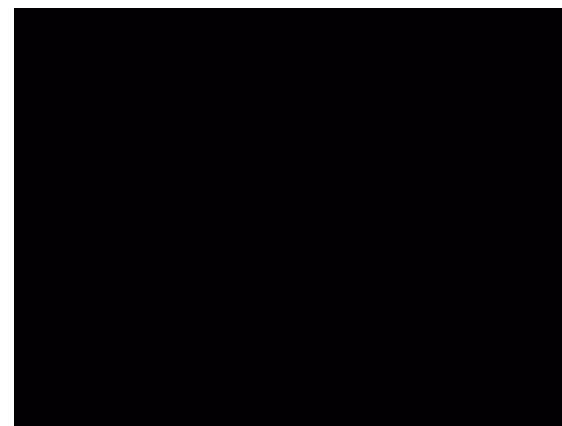
(Walter Kohn, Nobel Chemistry Prize, '98)



$$\psi(\mathbf{r}_1, \dots, \mathbf{r}_{N_{\text{el}}}) \quad O(C^N)$$



$$\{\psi_n(\mathbf{r}) | n = 1, \dots, N_{\text{el}}\} \quad O(N^3)$$



Constrained minimization problem:

Minimize:

$$E[\{\psi_n\}] = \sum_{n=1}^{N_{\text{el}}} \int d\mathbf{r} \psi_n^*(\mathbf{r}) \left(-\frac{\hbar^2}{2m_{\text{el}}} \frac{\partial^2}{\partial \mathbf{r}^2} + V_{\text{ion}}(\mathbf{r}) \right) \psi_n(\mathbf{r}) + \frac{e^2}{2} \iint d\mathbf{r} d\mathbf{r}' \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} + E_{\text{xc}}[\rho(\mathbf{r})]$$

with orthonormal constraints: $\int d^3r \psi_m^*(\mathbf{r}) \psi_n(\mathbf{r}) = \delta_{mn}$

$$\text{Charge density: } \rho(\mathbf{r}) = \sum_{n=1}^{N_{\text{el}}} |\psi_n(\mathbf{r})|^2$$



DFT: It's Algorithm

Density functional theory (DFT)

Hohenberg & Kohn, *Phys. Rev.* **136**, B864 ('64); W. Kohn [Nobel chemistry prize, '98]

$O(C^N)$ → $O(N^3)$
1 N -electron problem N 1-electron problems



Volume 140, Issue 18, 14 May 2014

SPECIAL TOPIC: ADVANCES IN DENSITY FUNCTIONAL THEORY

A divide-conquer-recombine algorithmic paradigm for large spatiotemporal quantum molecular dynamics simulations

Fuyuki Shimojo^{1,2}, Shinnosuke Hattori^{1,2}, Rajiv K. Kalia¹, Manaschai Kunaseth^{1,3}, Weiwei Mou¹, Aiichiro Nakano¹, Ken-ichi Nomura¹, Satoshi Ohmura^{1,2,4}, Pankaj Rajak¹, Kohei Shimamura^{1,2,5} and Priya Vashishta¹

+ VIEW AFFILIATIONS

J. Chem. Phys. **140**, 18A529 (2014); <http://dx.doi.org/10.1063/1.4869342>

$O(N^3)$ → $O(N)$
Mean-field theory Divide-conquer-recombine

Divide-Conquer-(Re)combine

- “The first was to never accept anything as true which I could not accept as obviously true.

The second was to divide each of the problems in as many parts as I should to solve them. The third, beginning with the simplest and easiest to understand matters, little by little, to the most complex knowledge.

And the last resolution was to make my enumerations so complete and my reviews so general that I could be assured that I had not omitted anything.” (René Descartes, *Discourse on Method*, 1637)

divide (conquer) recombine

- モデルの分割一再統合の方法の優れた点は、分割した要素的概念を、モデルの理解に役立つように再構成することができ、そこに創造の入り込む余地があるという点にある。」(福井謙一学問の創造、1987)

room for creativity

Kenichi Fukui [Nobel Chemistry Prize, '81]



Stochastic Model of Stock Prices

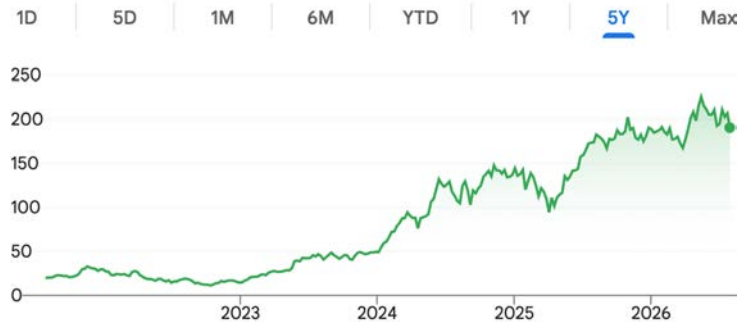
Fluctuation in stock price

NASDAQ: NVDA

190.01 USD ↑ 874.41% +170.51 past 5 years

Closed: Jul 29, 4:30 PM EDT • [Disclaimer](#)

After hours 189.06 -0.95 (0.50%)

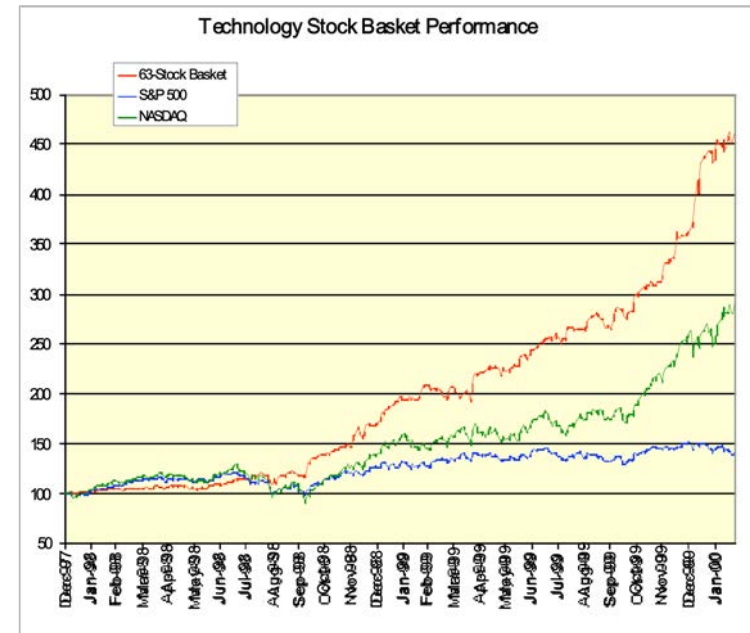


Basis of Black-Scholes analysis of option prices

$$dS = \mu S dt + \sigma S \varepsilon \sqrt{dt}$$

(1997 Nobel Economy Prize to Myron Scholes)

Computational stock portfolio trading



Andrey Omeltchenko ([Quantlab](#))

First Gauss Prize

The International Mathematical Union (IMU) and the Deutsche Mathematiker-Vereinigung (DMV) jointly award the Carl Friedrich Gauss Prize for Applications of Mathematics to **Professor Dr. Kiyoshi Itô**



for laying the foundations of the Theory of Stochastic Differential Equations and Stochastic Analysis. Itô's work has emerged as one of the major mathematical innovations of the 20th century and has found a wide range of applications outside of mathematics. Itô calculus has become a key tool in areas such as engineering (e.g., filtering, stability, and control in the presence of noise), physics (e.g., turbulence and conformal field theory), and biology (e.g., population dynamics). It is at present of particular importance in economics and finance with option pricing as a prime example.



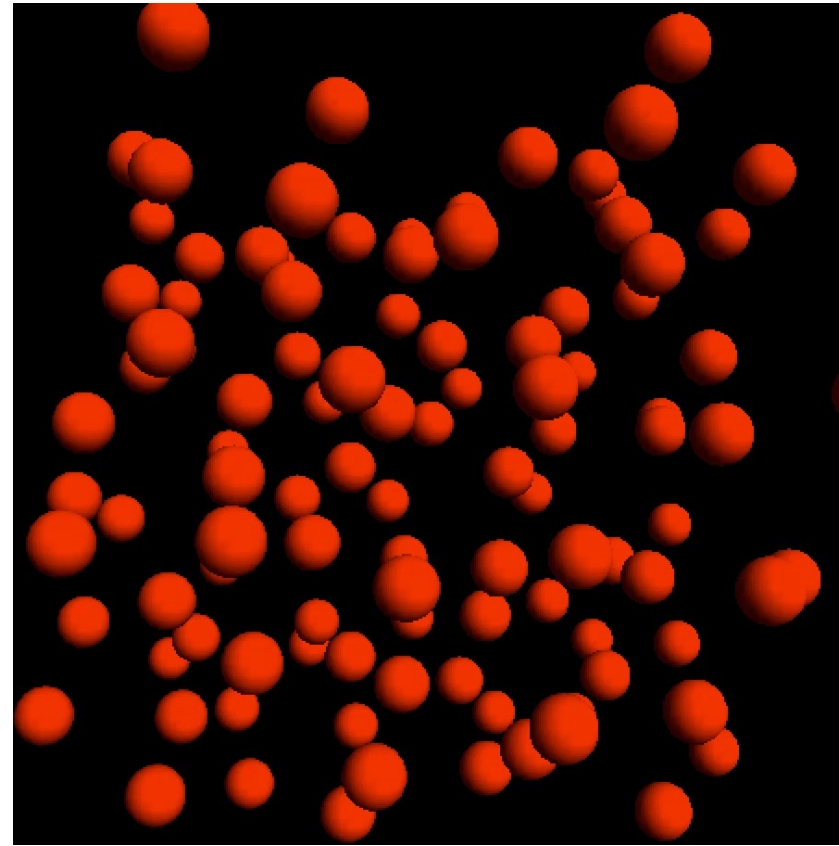
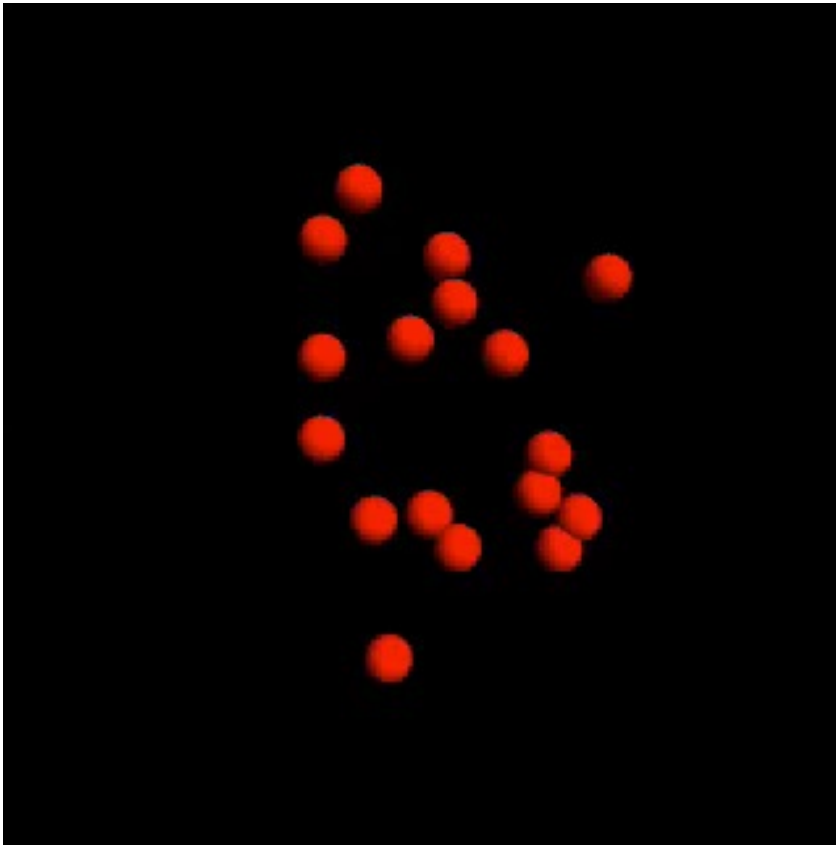
Madrid, August 22, 2006

Sir John Ball
President of IMU

$$dS = \mu S dt + \sigma S \varepsilon \sqrt{dt}$$

Günter M. Ziegler
President of DMV

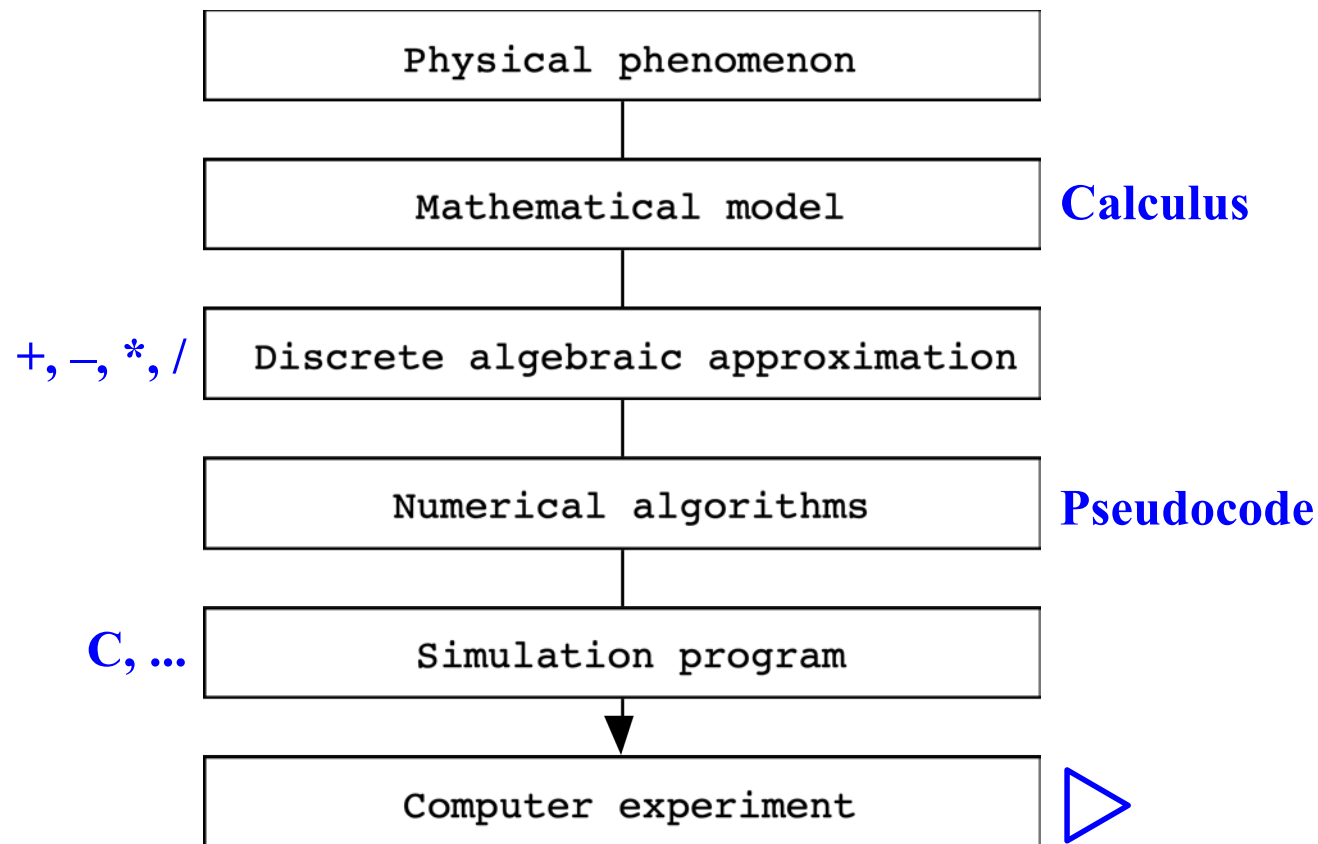
Monte Carlo Simulation



- **Random trial → acceptance by a cost criterion**
- **Combinatorial optimization by simulated annealing**

cf. Molecular dynamics

Recap: Scientific Computing



- Particle simulation as an archetypal example
- Use your own application in the final project

History of Particle Simulations

- '44 John von Neumann memo on a stored-program computer: *“Our present analytical methods seem unsuitable for the solution of the important problems arising in connection with **nonlinear partial differential equations**. The really efficient high-speed computing devices may provide us with those **heuristic hints** which are needed in all parts of mathematics for genuine progress”*
- '53 First Monte Carlo simulation of liquid by Metropolis, Rosenbluth, Rosenbluth, Teller, and Teller on MANIAC at Los Alamos Nat'l Lab
- '55 Enrico Fermi, John Pasta, and Stanislaw Ulam studied the dynamics of a one-dimensional array of particles coupled by anharmonic springs on MANIAC
- '56 Dynamics of hard spheres (billiards) studied by Alder and Wainwright at Lawrence Livermore Nat'l Lab.
- '60 Radiation damage in crystalline Cu studied with short-range repulsion and uniform attraction toward the center by George Vineyard's group at Brookhaven Nat'l Lab
- '64 First MD simulation of liquid (864 argon atoms) using interatomic potentials by Aneesur Rahman at Argonne Nat'l Lab on a CDC 3600

Simulation Pioneers in “Oppenheimer”



Benny Safdie
Edward Teller

Equation of State by Fast Computing Machines
Journal of Chemical Physics 21 (6), 1087-1093 (1953)
Received March 6, 1953; Published June 1953

NICHOLAS METROPOLIS
ARIANNA W. ROSENBLUTH
MARSHALL N. ROSENBLUTH
AUGUSTA H. TELLER
Los Alamos Scientific Laboratory

EDWARD TELLER
Department of Physics, University of Chicago

Who invented the algorithm...?

ARIANNA & MARSHALL ROSENBLUTH
not
Nicholas Metropolis

STUDIES OF NON LINEAR PROBLEMS

E. FERMI, J. PASTA, and S. ULAM
Document LA-1940 (May 1955).

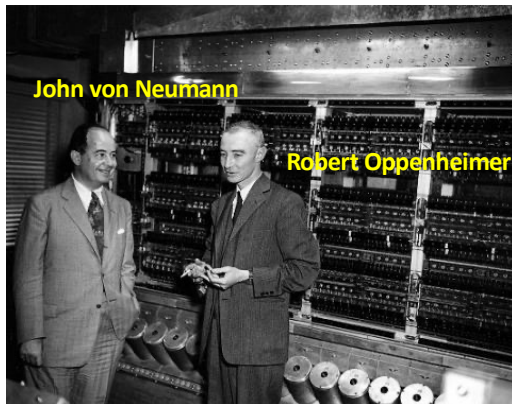
ABSTRACT.

A one-dimensional dynamical system of 64 particles with forces between neighbors containing nonlinear terms has been studied on the Los Alamos computer MANIAC I. The nonlinear terms considered are quadratic, cubic, and broken linear types. The results are analyzed into Fourier components and plotted as a function of time.



Danny Deferrari
Dr Enrico Fermi

And US Supercomputers



MANIAC computer at Los Alamos



Aichiro Nakano of the University of Southern California uses the Argonne Leadership Computing Facility to study how light can change the structures and properties of atomically thin materials to create better, cheaper semiconductors. (Image by Argonne National Laboratory.)

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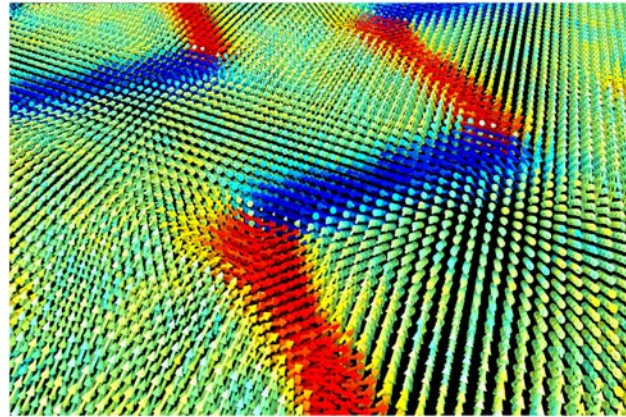
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Quanta in bulk

A USC computer scientist wants to produce quantum materials at scale, with help from Argonne supercomputers.



Topological structure during ferroelectric switching in PbTiO_3 , where arrows represent electrical polarization colored according to its magnitude. Image courtesy of Thomas Linker and Ken-ichi Nomura/USC Viterbi School of Engineering.

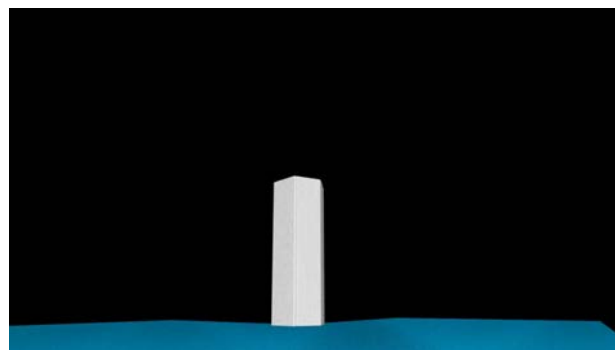
Supercomputing at Argonne National Laboratory now

Simulation to World Model

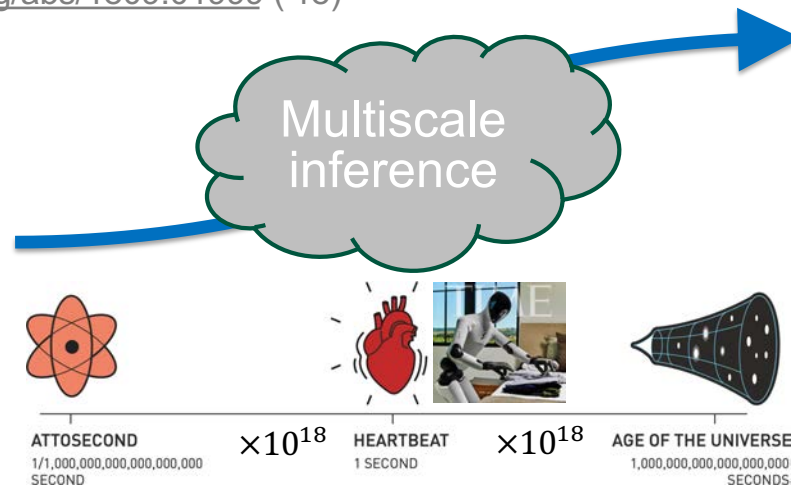
- **Simulation (computational science):** Third branch of science—complements theory & experiment (Manhattan project, 1940's-)
- **Digital twin:** Simulation + dynamic real-world data (NASA missions; 1960's-)
Example: Prof. Chenfanfu Jiang (UCLA)
https://drive.google.com/file/d/14CxaCBuYkZox9hcFeCsgM_3N5oobmuWf/view
- **World model:** Digital twin + learned prediction (AI accelerator; 2020's-) David Ha *et al.*, <https://arxiv.org/abs/1809.01999> ('18)



Macroscopic world



Atomistic-scale simulation



Power of Scale

- **Allegro-FM (foundation model) for NNQMD** described material properties & processes accurately, covering 89 elements in the periodic table, exhibiting *emergent capabilities* for which the model was not trained Nomura *et al.*, *J. Phys. Chem. Lett.* **16**, 6637 ('25)
- **E3D: Allegro-FM trained *at scale* learned interpretable chemical-bond *concepts* without supervision** Hattori *et al.*, *Nature Commun.* doi: 10.1038/s41467-026-74919-8 ('26)
E3D: Edge-wise emergent energy-decomposition framework ~ computational imaging to peek into AI brain



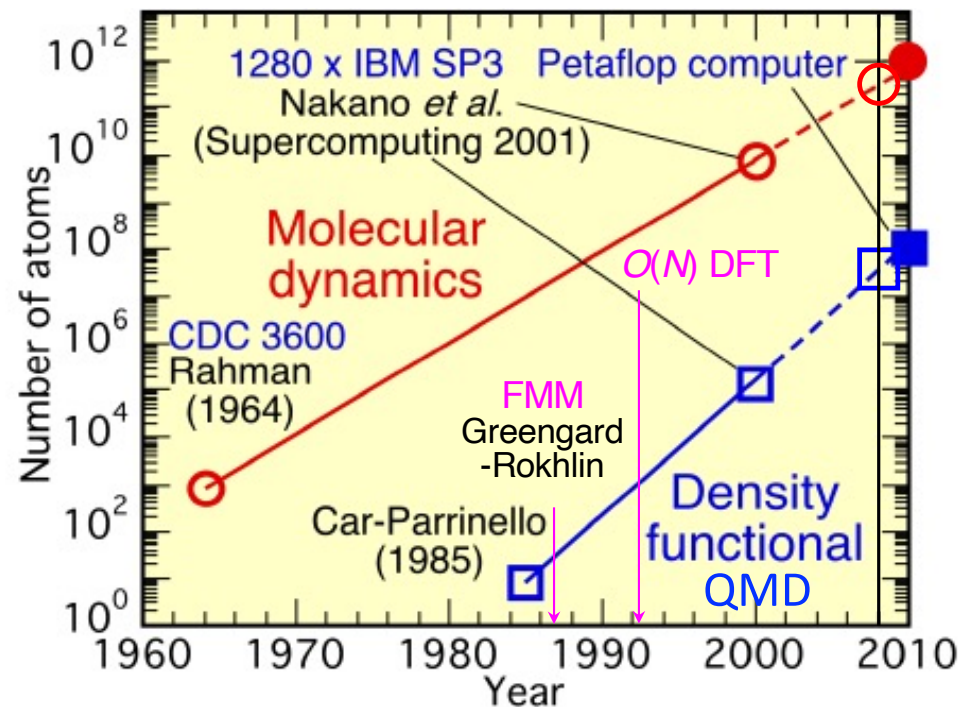
P. W. Anderson, *Science* **177**, 393 ('72)
Nobel physics prize ('77)



Moore's Law in Scientific Computing

Number of particles in MD simulations has doubled:

- Every 19 months in 50 years for classical MD
- Every 22 months in 30 years for quantum MD



10^{12} -atom MD & 10^8 -electron QMD on a petaflop/s computer with advances in algorithmic & parallel-computing techniques

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It's algorithm!

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1987 – Leslie Greengard
Series Winner (1987)
Citation
For his dissertation "The Rapid Evaluation of Potential Fields in Particle Systems." $O(N^2) \rightarrow O(N)$



Top 10 Algorithms in History

- Metropolis Algorithm for Monte Carlo
- Simplex Method for Linear Programming
- Krylov Subspace Iteration Methods
- The Decompositional Approach to Matrix Computations
- The Fortran Optimizing Compiler
- QR Algorithm for Computing Eigenvalues
- Quicksort Algorithm for Sorting
- Fast Fourier Transform
- Integer Relation Detection
- Fast Multipole Method

IEEE CiSE, Jan/Feb ('00)

<http://awards.acm.org/doctoral%5Fdissertation>

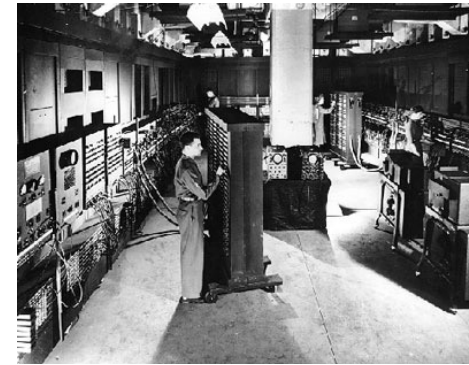
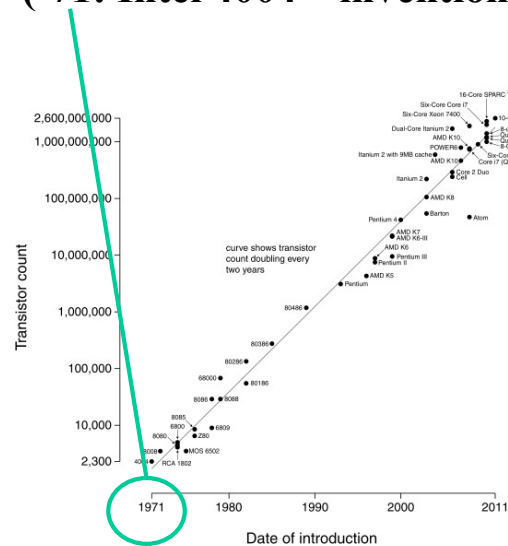
History of Supercomputers

Early '40s: ENIAC by Presper Eckert & John Mauchly at Univ. of Pennsylvania—first general-purpose electronic computer

'76: Cray 1 by Seymour Cray—beginning of vector supercomputer era

Late 80's: massively parallel computers such as the Thinking Machines CM-2

(’71: Intel 4004—invention of microprocessor)



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1985 – Daniel Hillis
Series Winner (1985)
Citation
For his dissertation "The Connection Machine."

New Cancer Research Center to be Based at USC

By James Grant on October 26, 2009 7:55 AM

USC has been selected to establish a \$16 million cancer research center as part of a new strategy against the disease by the U.S. National Institutes of Health and its National Cancer Institute.

The five-year award will create a National Cancer Institute Physical Science-Oncology Center based at USC and involving a consortium of universities. Partnering in the USC grant will be Arizona State University, the California Institute of Technology, Cold Spring Harbor Laboratory, New York University, Stanford University, the University of Arizona and the University of Texas at Austin.

The Physical Science-Oncology Center initiative differs from past cancer research programs. While cancer biologists often work with scientists in other fields, this marks the first large-scale recruitment of outside scientists in the battle against the disease.



USC Viterbi School of Engineering
professor and principal investigator
W. Daniel Hillis

[E-MAIL](#) [PRINT](#) [SHARE](#)

Digress: Birth of FORTRAN Programming



N.S. Scott et al.,
Comput. Phys. Commun.
252, 107269 (2020)

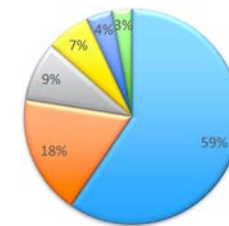
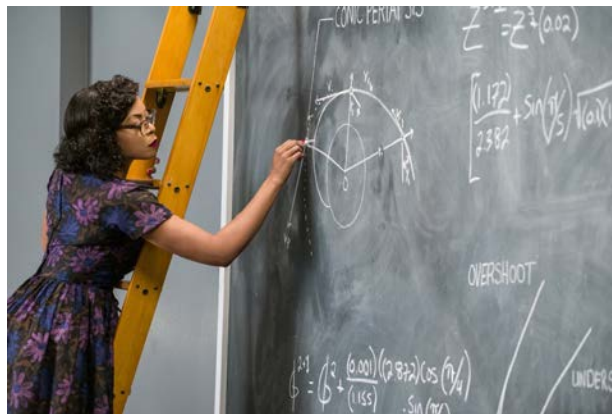


Fig. 13. The range of programming languages used across the Program Library's 3224 published programs (1969–2016).



Math
vs.
Computing



cf. <https://aiichironakano.github.io/cs596/Perkel-ScienceCode-NPhys21.pdf>

Merge of PC & Supercomputers

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	LineShine - LingKun, LX2 304C 1.55GHz, LingQi, Kylin OS, Shenzhen Cloud Computing Center Co., Ltd. National Supercomputing Centre in Shenzhen (NSCS) China	13,789,440	2,198.40	2,735.82	42,220
2	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
3	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot- 11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
4	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698

Theoretical performance

Measured performance (Pflop/s)

**Flop/s =
floating-point
operations/second**

M (mega) = 10^6

G (giga) = 10^9

T (Tera) = 10^{12}

P (Peta) = 10^{15}

X (Exa) = 10^{18}

<http://www.top500.org> (June '26)

- **USC-CARC: 20,000 cores**
- **CACS: 4,096 cores**
- **CACS-INCITE: 1M node-hrs/yr on Exaflop/s Aurora at Argonne**

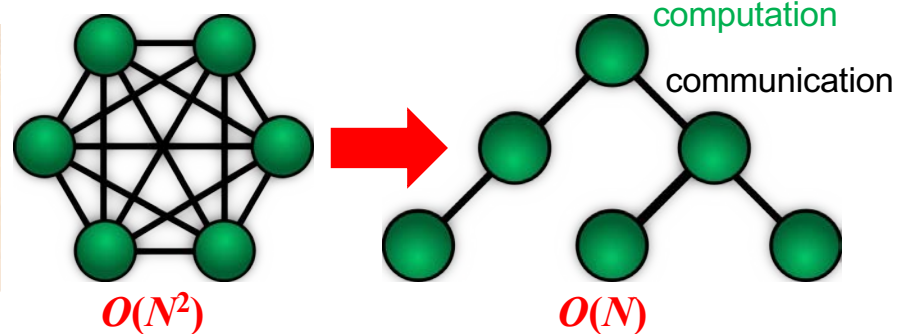
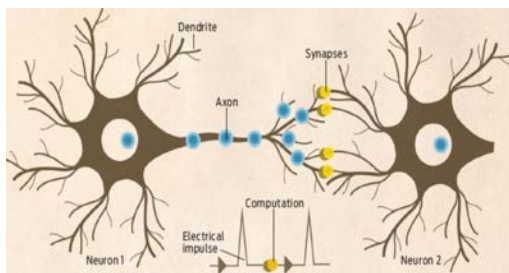


Why Supercomputer?

- Q.** If supercomputers use the same processors as laptops & cell phones, could we just use cloud-computing resources to do the same?
- A.** No. Emergent properties arising from many interacting entities (e.g., life) could only be studied as a whole [**More is different**,” P. W. Anderson, *Science* **177**, 393 ('72) <https://aiichironakano.github.io/cs596/Anderson-MoreIsDifferent-Science72.pdf>]. This would require innovative hardware & software solutions to tightly couple massive computations:

Hardware: Ultrafast network interconnect among billions of processing elements, *cf.* human brain has 10^3 synapses per neuron.

Software: Communication-minimizing divide-conquer-recombine algorithms that accurately describe many-body interaction.



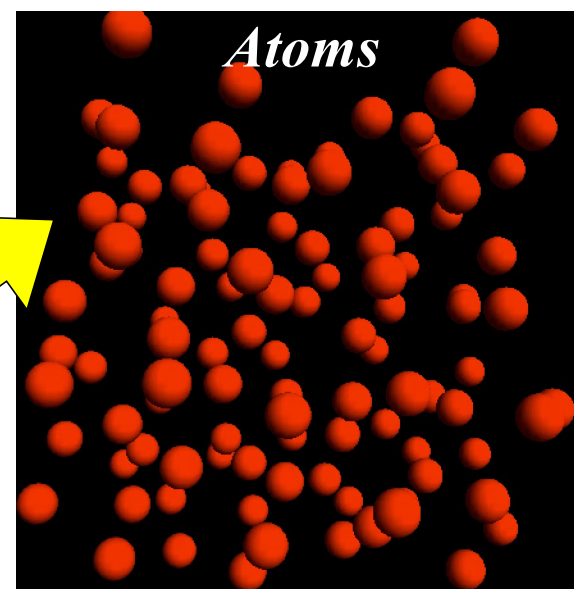
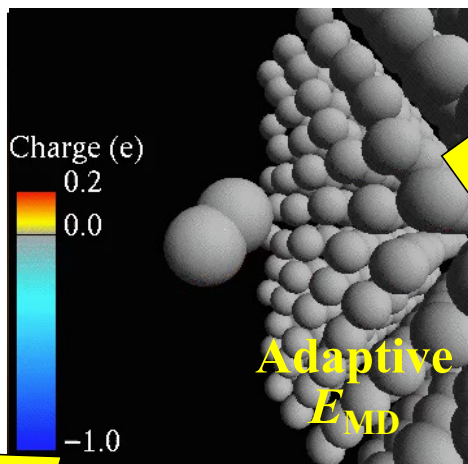
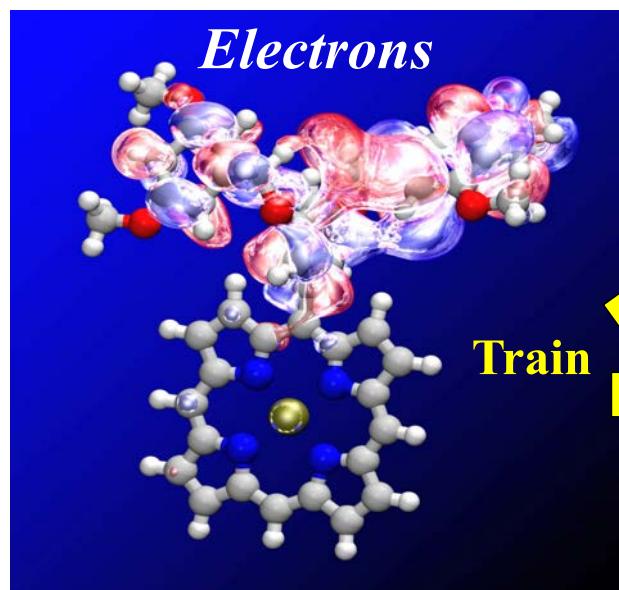
Molecular Dynamics Simulation & Machine Learning

*At the postExa/quantum/AI
nexus*

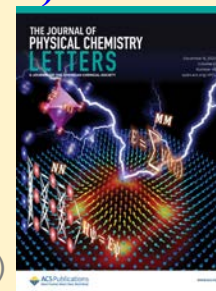
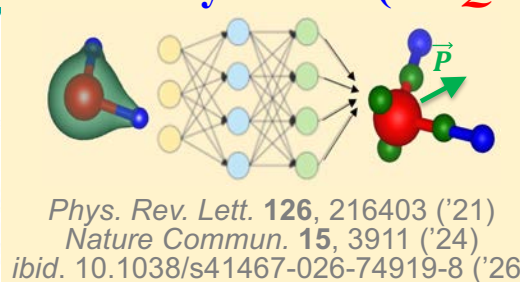
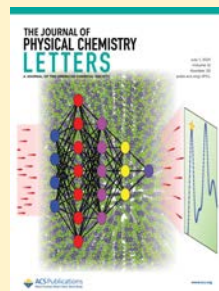
Molecular Dynamics (*MD*)

Reactive MD (*RMD*)

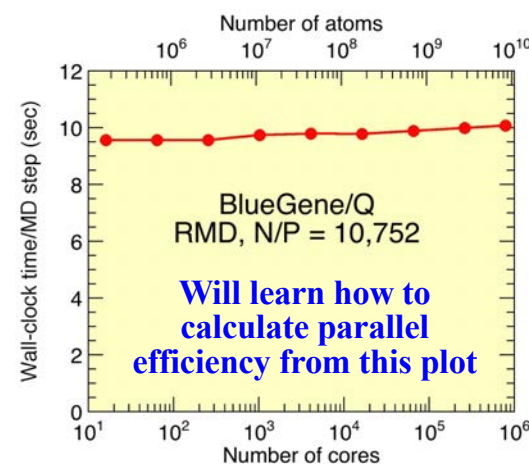
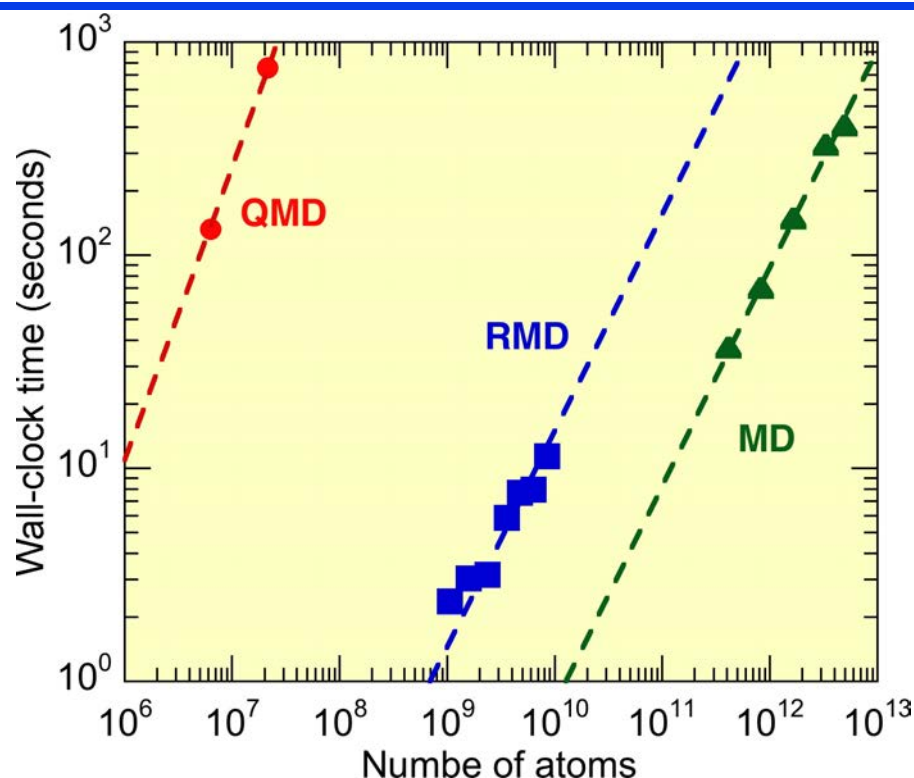
Nonadiabatic quantum
MD (*NAQMD*)



First principles-based neural-network quantum
molecular dynamics (*NNQMD*)



Scalable Simulation Algorithm Suite



QMD (quantum molecular dynamics): DC-DFT

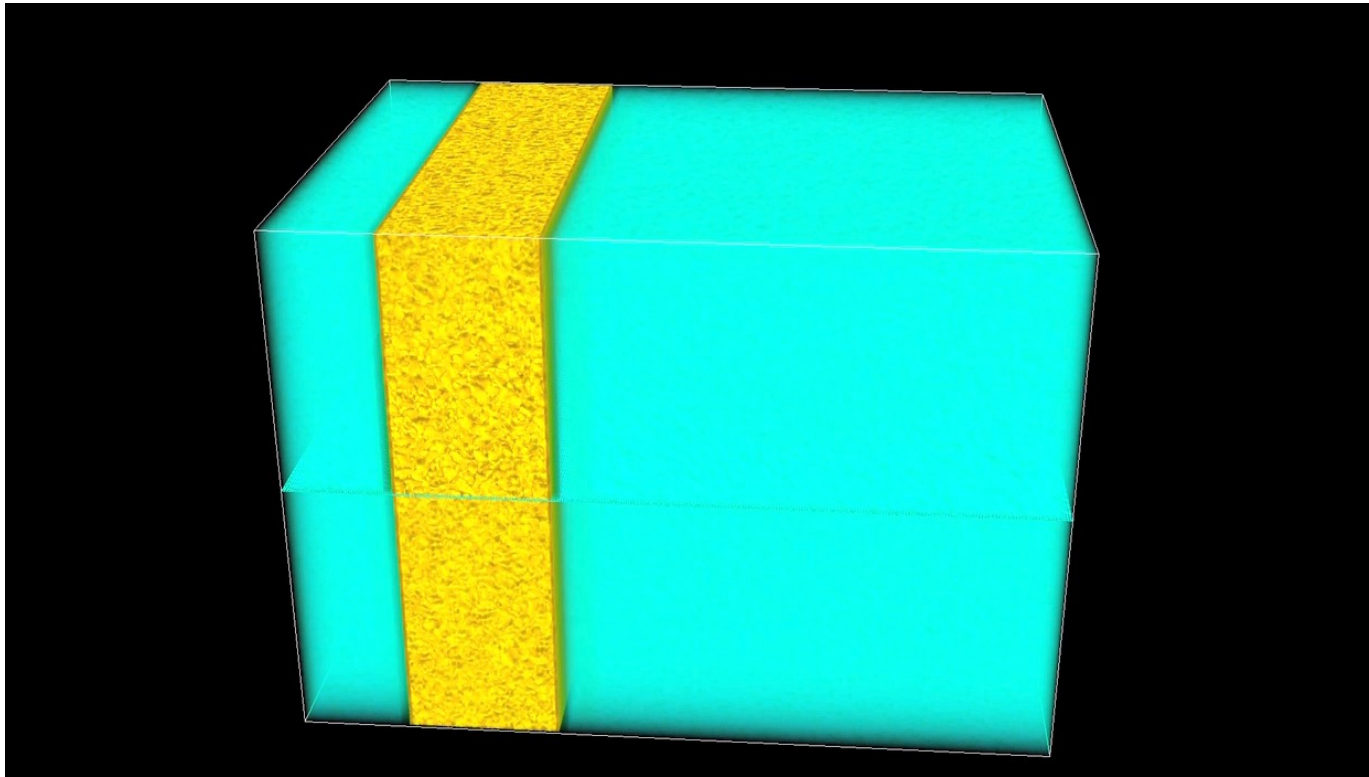
RMD (reactive molecular dynamics): F-ReaxFF

MD (molecular dynamics): MRMD

- 4.9 trillion-atom space-time multiresolution MD (MRMD) of SiO_2
 - 8.5 billion-atom fast reactive force-field (F-ReaxFF) RMD of RDX
 - 39.8 trillion grid points (50.3 million-atom) DC-DFT QMD of SiC
- parallel efficiency over 0.98 on 786,432 Blue Gene/Q cores

Nanobubble Collapse Near Silica Surface

- **Billion-atom MD simulation of shock-induced nanobubble collapse in water near silica surface**
(67 million core-hours on 163,840 Blue Gene/P cores)

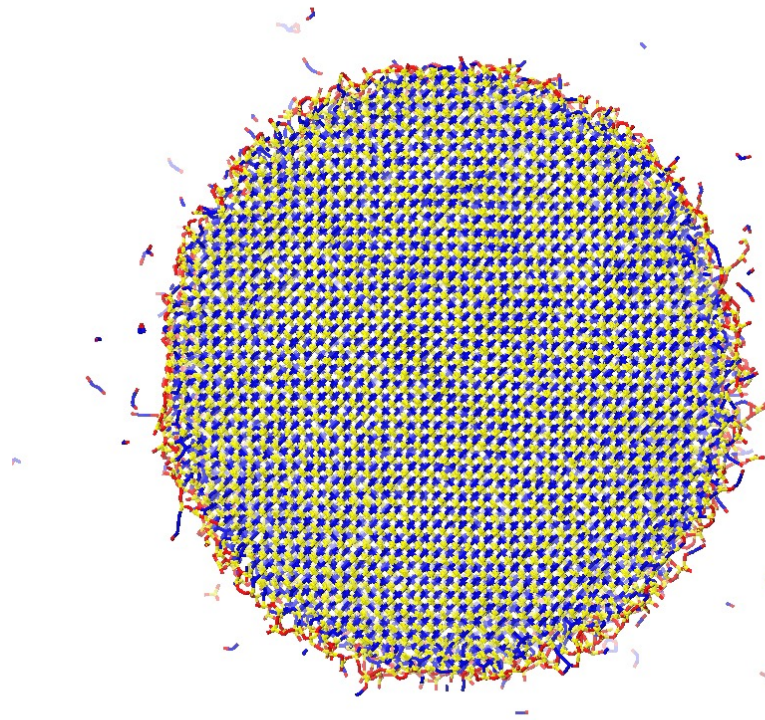


- **Water nanojet formation and its collision with silica surface**

A. Shekhar *et al.*, *Phys. Rev. Lett.* **111**, 184503 ('13)

Novel Nano-carbon Synthesis

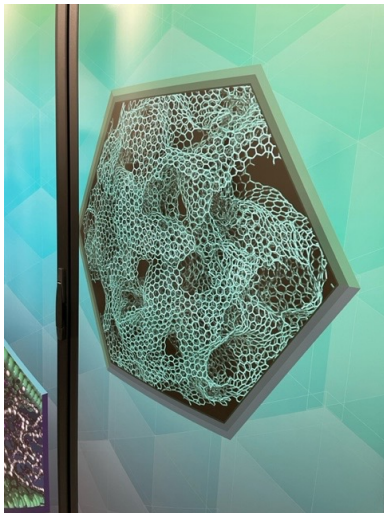
112 million-atom reactive molecular dynamics (RMD) simulation of high-temperature oxidation of SiC nanoparticle on 786,432 IBM Blue Gene/Q cores



K. Nomura *et al.*, *Sci. Rep.* **6**, 24109 ('16)

Argonne National Laboratory and Hewlett Packard Enterprise prepare for exascale era with new testbed supercomputer

AUTHOR ARGONNE NATIONAL LABORATORY AND
HEWLETT PACKARD ENTERPRISE
PUBLISHED 08/25/2021



Argonne Theta

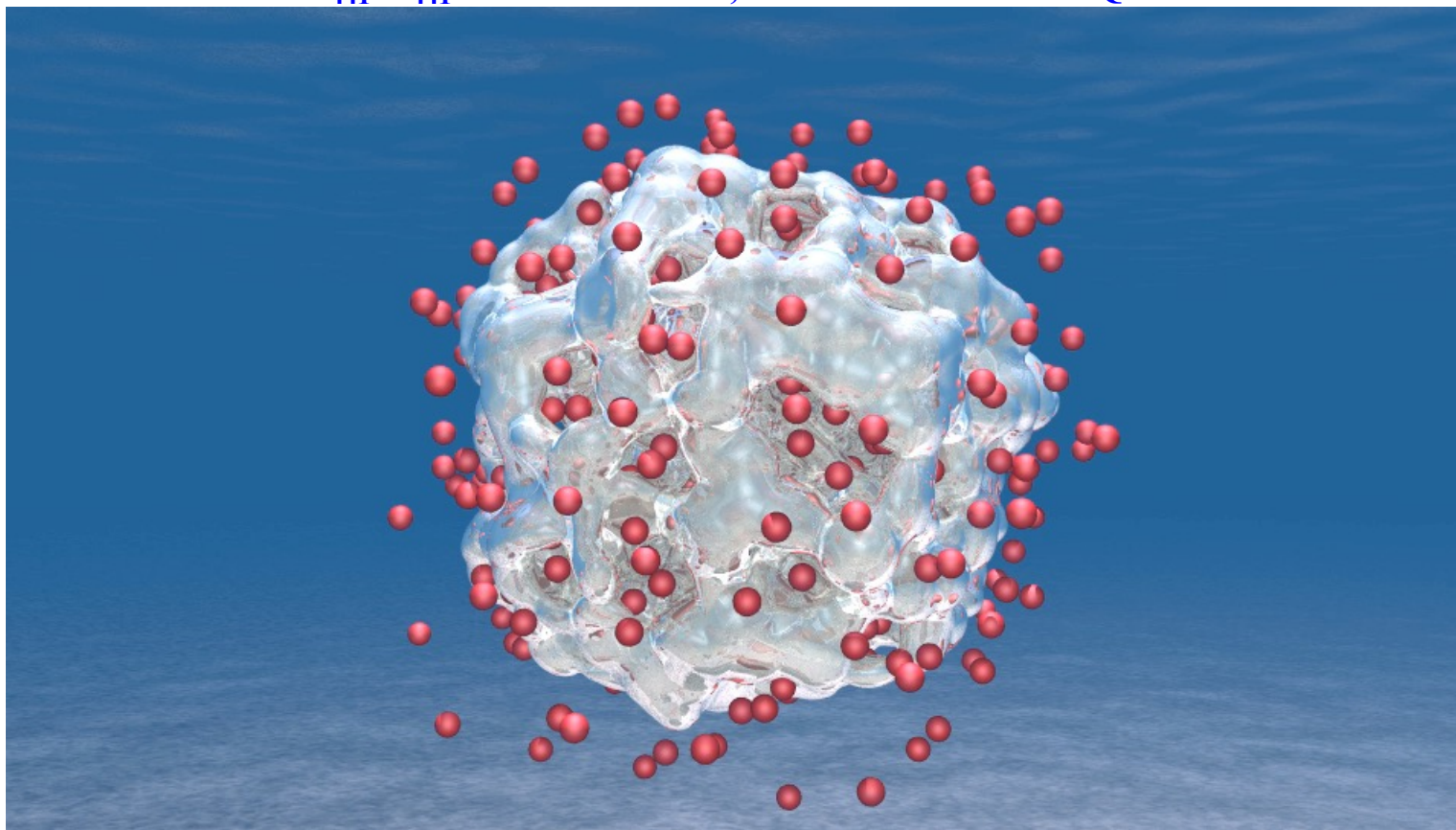


Argonne National Laboratory's new system, Polaris, built by HPE, will optimize AI, engineering and scientific projects for the forthcoming Aurora exascale supercomputer.

Argonne News (Aug. 25, 2021)

H₂ Production from Water Using LiAl Particles

16,661-atom quantum molecular dynamics (QMD) simulation
of Li₄₄₁Al₄₄₁ in water on 786,432 IBM Blue Gene/Q cores



K. Shimamura *et al.*, *Nano Lett.* **14**, 4090 ('14); K. Nomura *et al.*, SC14 ('14)

Enabling Computer Science Technologies

What can computer science do to enable extreme-scale computational science?

A lot! That's what you will learn in CSCI 596:

- **Parallel computing**
- **Visualization**
- **Grid/cloud computing**

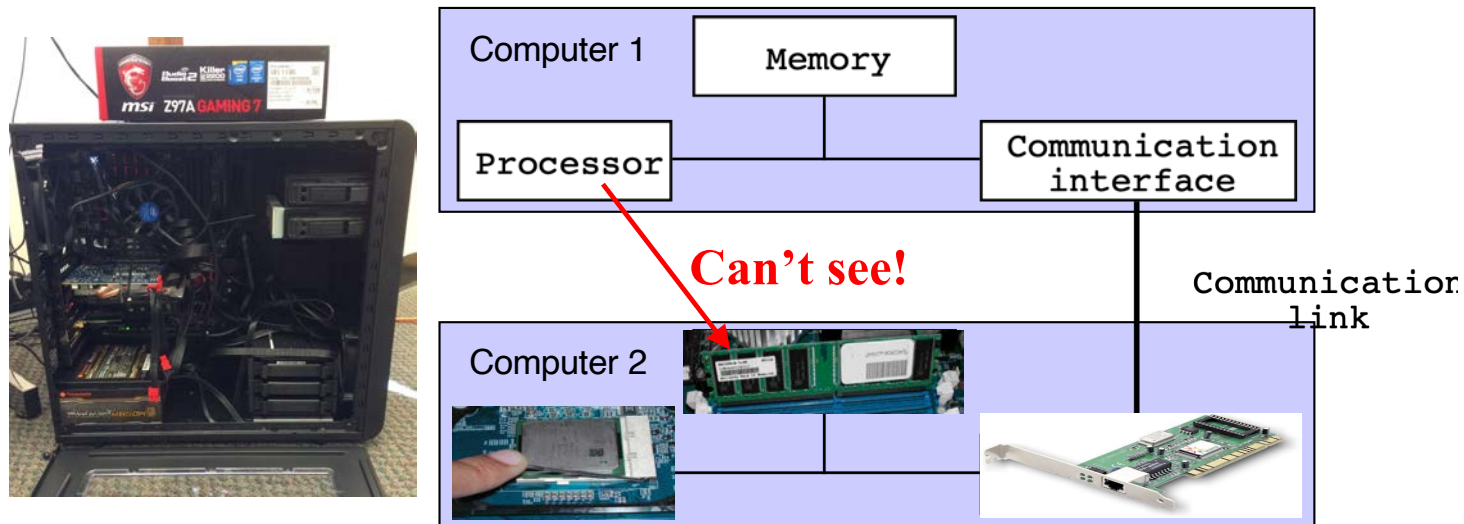


Parallel Computing

Parallel MD algorithm: Easy!

1. $\vec{v}_i(t + \frac{\Delta}{2}) \leftarrow \vec{v}_i(t) + \frac{\Delta}{2} \vec{a}_i(t)$
2. $\vec{r}_i(t + \Delta) \leftarrow \vec{r}_i(t) + \vec{v}_i(t + \frac{\Delta}{2})\Delta$
3. `atom_migrate()`
4. `atom_cache()`
5. Compute $\vec{a}_i(t + \Delta)$ as a function of $\{\vec{r}_i(t + \Delta)\}$
6. $\vec{v}_i(t + \Delta) \leftarrow \vec{v}_i(t + \frac{\Delta}{2}) + \frac{\Delta}{2} \vec{a}_i(t + \Delta)$

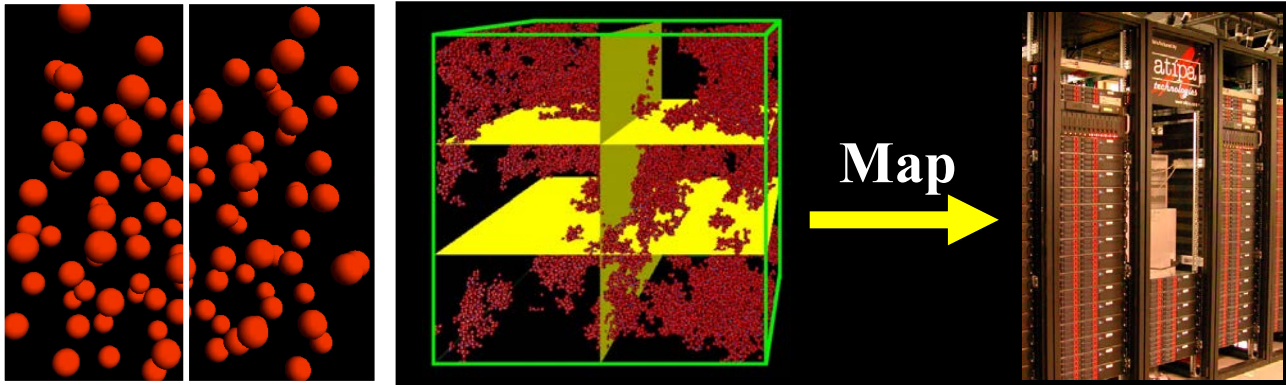
Parallel Computing Hardware



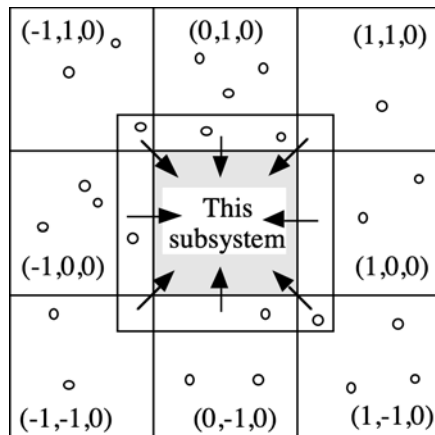
- **Processor:** Executes arithmetic & logic operations
- **Memory:** Stores program & data (**stored program computer**)
- **Communication interface:** Performs signal conversion & synchronization between communication link & a computer
- **Communication link:** A wire capable of carrying a sequence of bits as electrical (or optical) signals

Parallel Molecular Dynamics

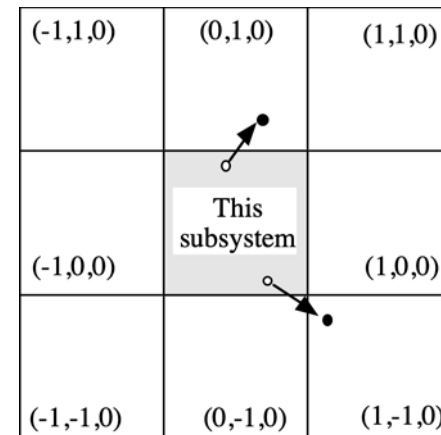
Spatial decomposition (short ranged): $O(N/P)$ computation



Atom caching: $O((N/P)^{2/3})$



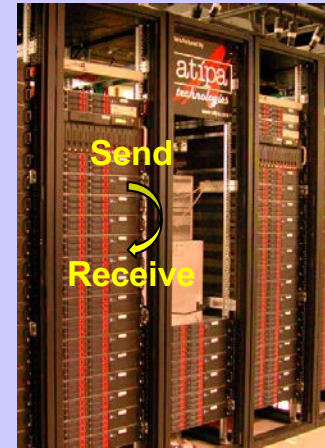
Atom migration



MPI Programming

```
#include "mpi.h"
#include <stdio.h>
main(int argc, char *argv[]) {
    MPI_Status status;
    int myid;
    int n;
    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &myid);
    if (myid == 0) {
        n = 777;
        MPI_Send(&n, 1, MPI_INT, 1, 10, MPI_COMM_WORLD);
    }
    else {
        MPI_Recv(&n, 1, MPI_INT, 0, 10, MPI_COMM_WORLD, &status);
        printf("n = %d\n", n);
    }
    MPI_Finalize();
}
```

Only need two (send & receive) functions!

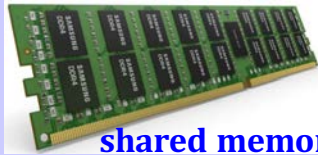


<https://www.mcs.anl.gov/research/projects/mpi>

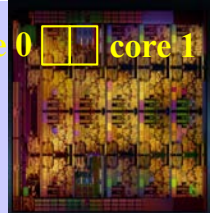
OpenMP Programming

```
#include <stdio.h>
#include <omp.h>
void main () {
    int nthreads,tid;
    nthreads = omp_get_num_threads();
    printf("Sequential section: # of threads = %d\n",nthreads);
    /* Fork multi-threads with own copies of variable */
    #pragma omp parallel private(tid)
    {
        /* Obtain & print thread id */
        tid = omp_get_thread_num();
        printf("Parallel section: Hello world from thread %d\n",tid);
        /* Only master thread does this */
        if (tid == 0) {
            nthreads = omp_get_num_threads();
            printf("Parallel section: # of threads = %d\n",nthreads);}
    } /* All created threads terminate */
}
```

parallel section



shared memory



core 0 core 1

- Obtain the number of threads & my thread ID
- By default, all variables are shared unless selectively changing storage attributes using private clauses

<http://www.openmp.org>

GPU Programming: CUDA

- **Compute Unified Device Architecture**
- **Integrated host (CPU) + device (GPU) application programming interface based on C language developed at NVIDIA**

- **CUDA homepage**

`http://www.nvidia.com/object/cuda\_home\_new.html`

- **Compilation**

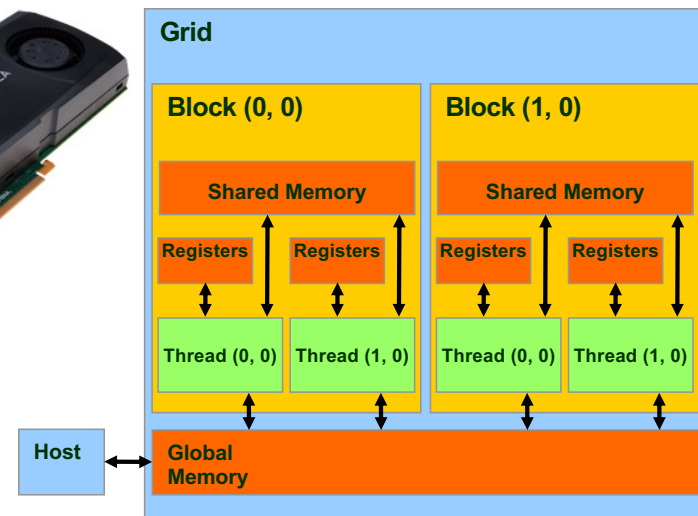
`$ nvcc pi.cu`

- **Execution**

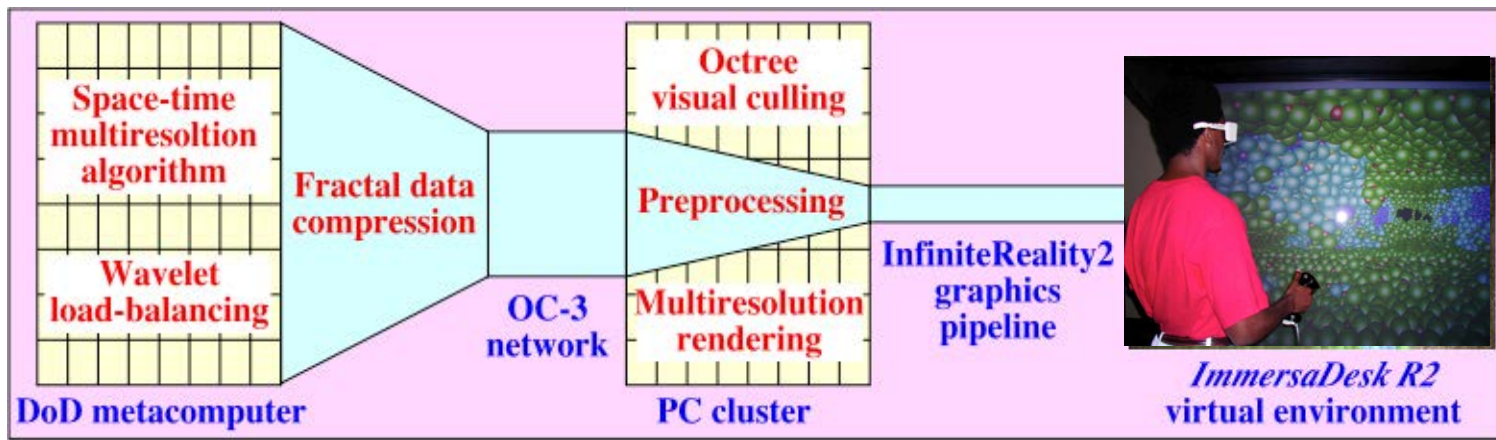
`$ a.out`

PI = 3.141593

**New ways: OpenMP target offload
& SYCL (see [link](#))**



Immersive & Interactive Visualization

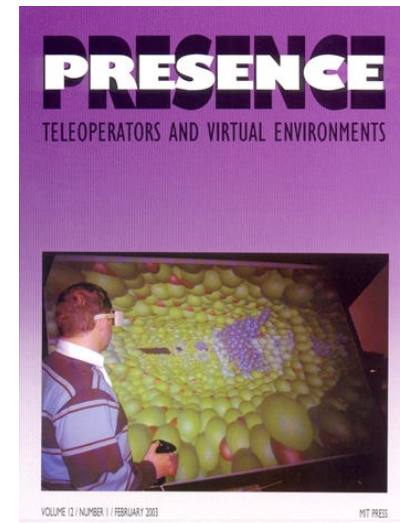


Basics:

- **OpenGL** programming/CAVE library

Billion-atom walkthrough:

- Octree-based view frustum culling
- Probabilistic occlusion culling
- Multiresolution rendering
- Parallel & distributed processing



OpenGL Programming

```
...
#include <OpenGL/gl.h>
#include <OpenGL/glu.h>
#include <GLUT/glut.h>
...
/* Set a glut callback function */
glutDisplayFunc(display);

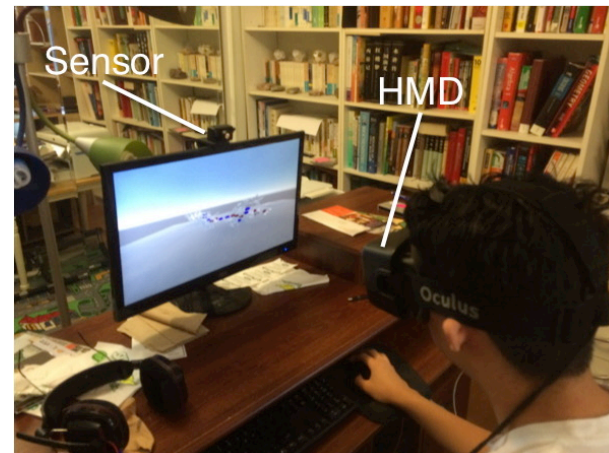
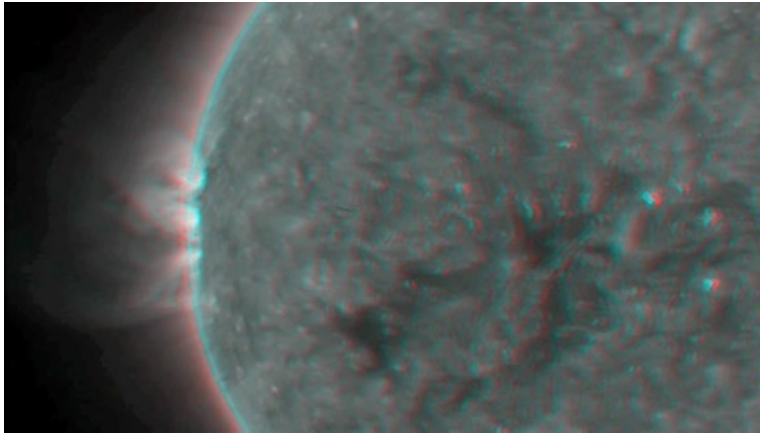
/* generate an OpenGL display list for single sphere */
sphereid = glGenLists(1);
makeFastNiceSphere(sphereid,atom_radius);

/* generate an OpenGL display list for the atoms' geometry */
atomsid = glGenLists(1);
/* make the geometry of the current frame's atoms */
makeCurframeGeom();

/* Start main display loop */
glutMainLoop();
```

<http://www.opengl.org>

3D Is Back in Hollywood (and Home)

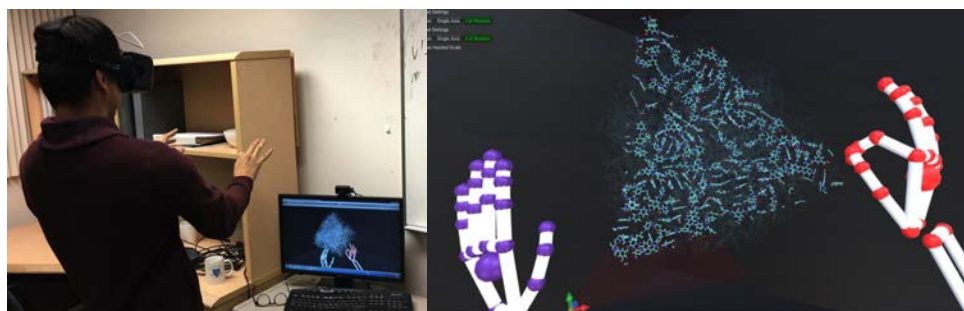


<https://github.com/USCCACS/GEARS>

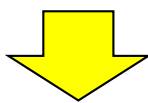


Oculus Rift & HTC Vive head mounted displays

Scientific Mixed Reality



VR



MR



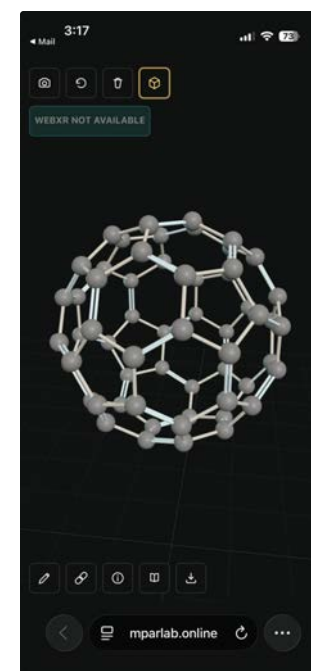
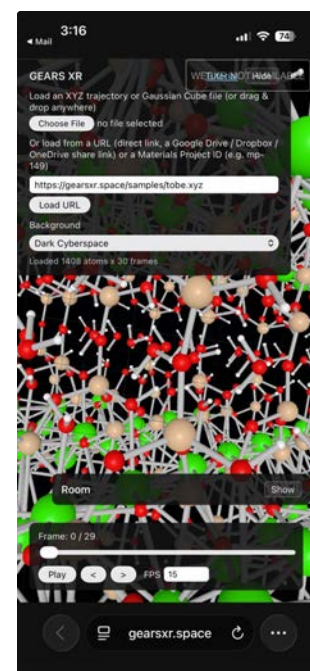
Microsoft Hololens academic seeding program at USC

cf. CSCI 538: Augmented, Virtual and Mixed Reality

AI-generated MR app

<https://gearsxr.space/>

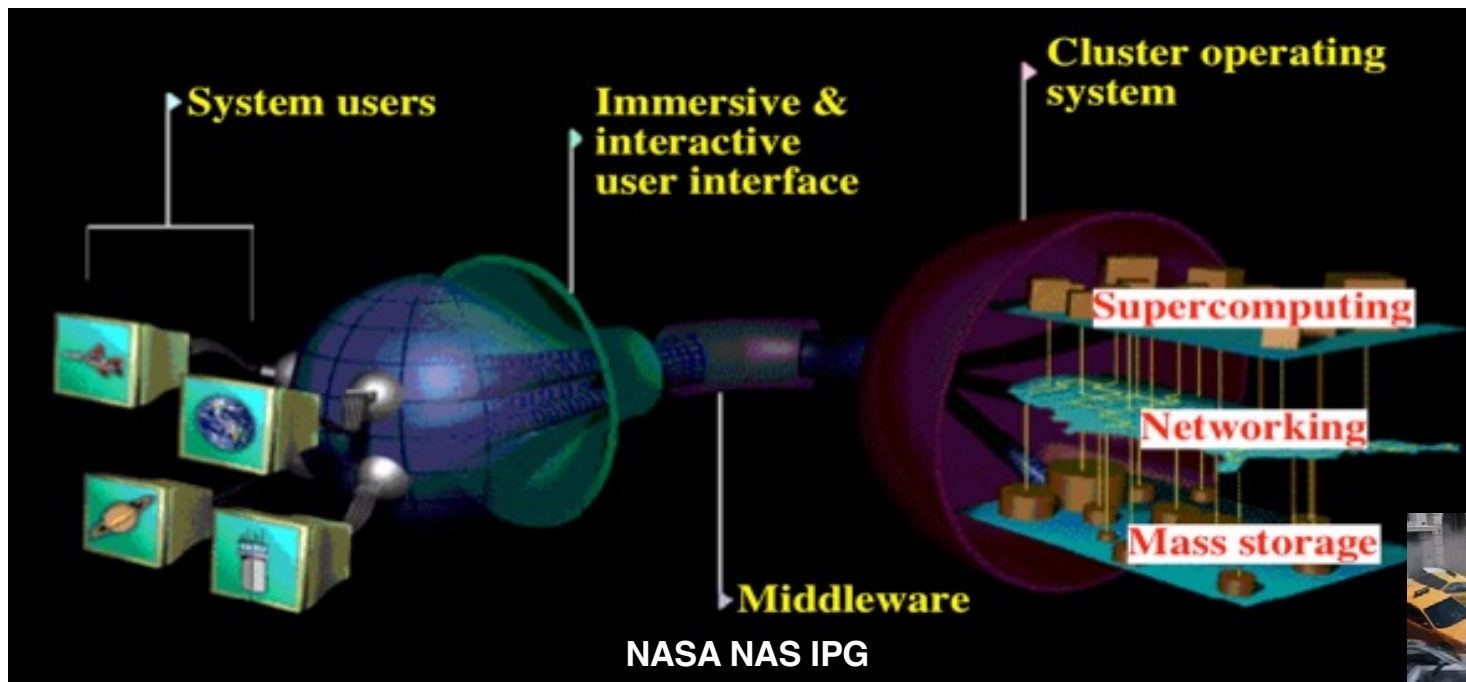
<https://kenichinomura.github.io/PolymerARLab/>



Final project?

Grid Computing

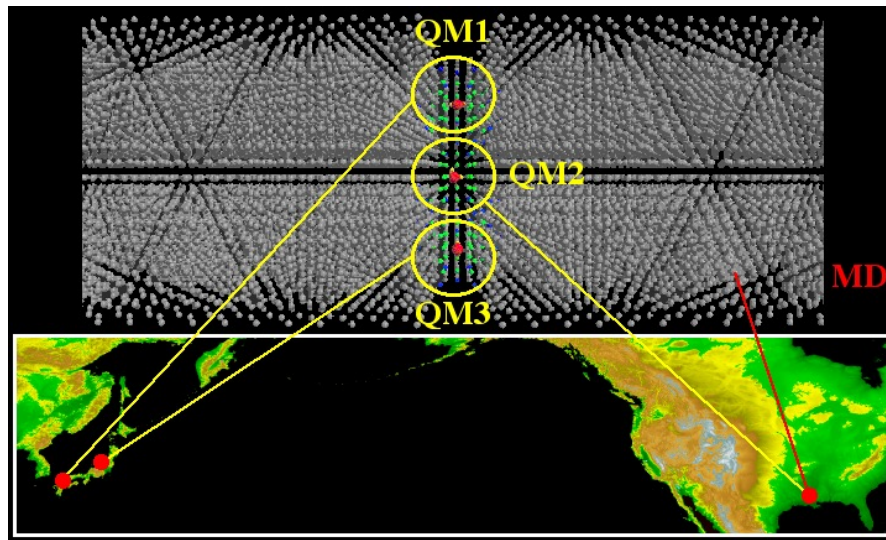
- **World Wide Web:** Universal interface to digital library on the Internet
- **Information Grid:** Pervasive (from any place in the world at any time) access to everything (computing, storage, experimental equipments, sensors, *etc.*, on high-speed networks)



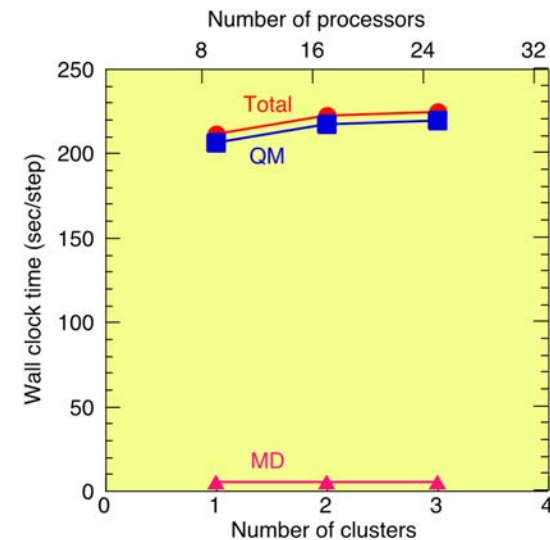
Global Collaborative Simulation

Multiscale MD/QM simulation on a Grid of distributed PC clusters in the US & Japan

- Task decomposition (MPI Communicator) + spatial decomposition
- MPICH-G2/Globus



Japan: Yamaguchi—65 P4 2.0GHz
Hiroshima, Okayama, Niigata—3×24 P4 1.8GHz
US: Louisiana—17 Athlon XP 1900+



MD — 91,256 atoms
QM (DFT) — 76*n* atoms on *n* clusters

- Scaled speedup, $P = 1$ (for MD) + $8n$ (for QM)
- Efficiency = 94.0% on 25 processors over 3 PC clusters

Kikuchi *et al.*, *IEEE/ACM SC02*

Multiscale Modeling

The Nobel Prize in Chemistry 2013



© Nobel Media AB
Martin Karplus



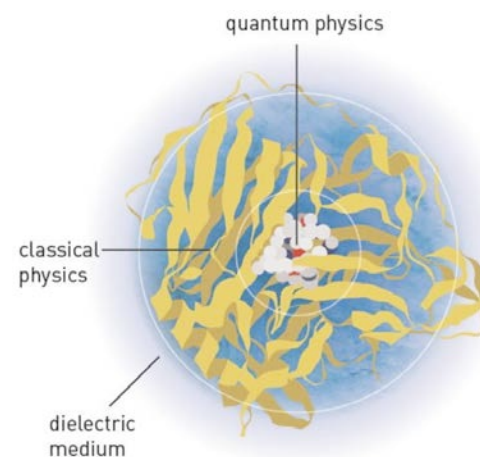
Photo: Keilana via
Wikimedia Commons
Michael Levitt



Photo: Wikimedia
Commons
Arieh Warshel

The Nobel Prize in Chemistry 2013 was awarded jointly to Martin Karplus, Michael Levitt and Arieh Warshel *"for the development of multiscale models for complex chemical systems"*.

A. Warshel & M. Karplus, *J. Am. Chem. Soc.* **94**, 5612 ('72)
A. Warshel & M. Levitt, *J. Mol. Biol.* **103**, 227 ('76)

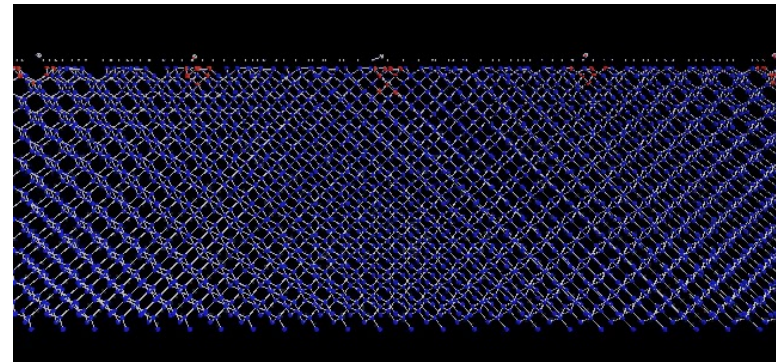
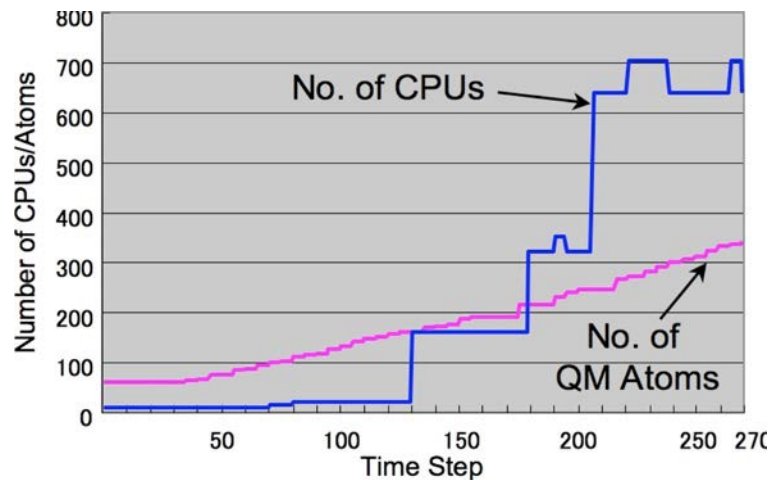


QM/MM:
**quantum-mechanical/
molecular-mechanical
modeling**

Sustainable Grid Supercomputing

- Sustained (> months) supercomputing (> 10^3 CPUs) on a Grid of geographically distributed supercomputers
- Hybrid Grid remote procedure call (GridRPC) + message passing (MPI) programming
- Dynamic allocation of computing resources on demand & automated migration due to reservation schedule & faults

Ninf-G GridRPC: ninf.apgrid.org; MPICH: www.mcs.anl.gov/mpi



Takemiya *et al.*, *IEEE/ACM SC06*
Song *et al.*, *IJCS* ('09)

Multiscale QM/MD simulation of high-energy beam oxidation of Si

Above the Clouds: A Berkeley View of Cloud Computing



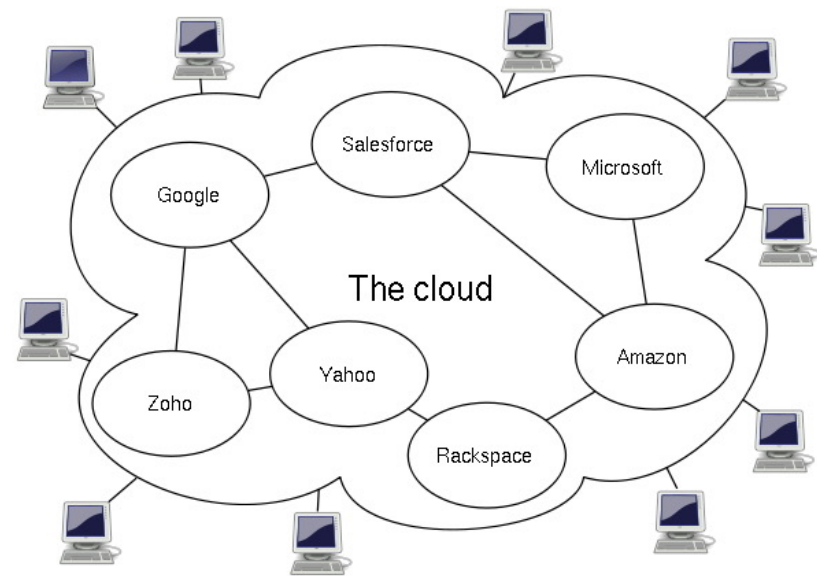
*Michael Armbrust
Armando Fox
Rean Griffith
Anthony D. Joseph
Randy H. Katz
Andrew Konwinski
Gunho Lee
David A. Patterson
Ariel Rabkin
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Technical Report No. UCB/EECS-2009-28
<http://www.eecs.berkeley.edu/Pubs/TechRpts/2009/EECS-2009-28.html>

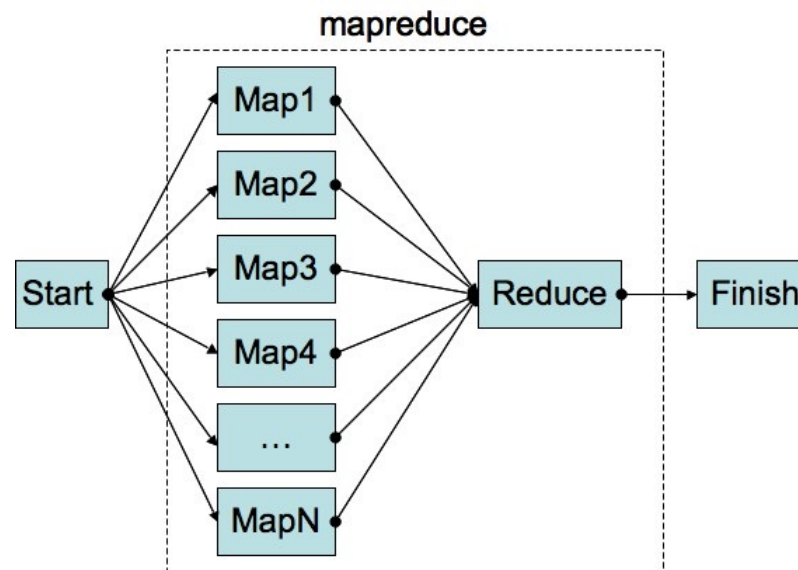
Cloud computing at CARC

<https://www.carc.usc.edu/user-guides/hpc-systems/artemis/artemis-overview>

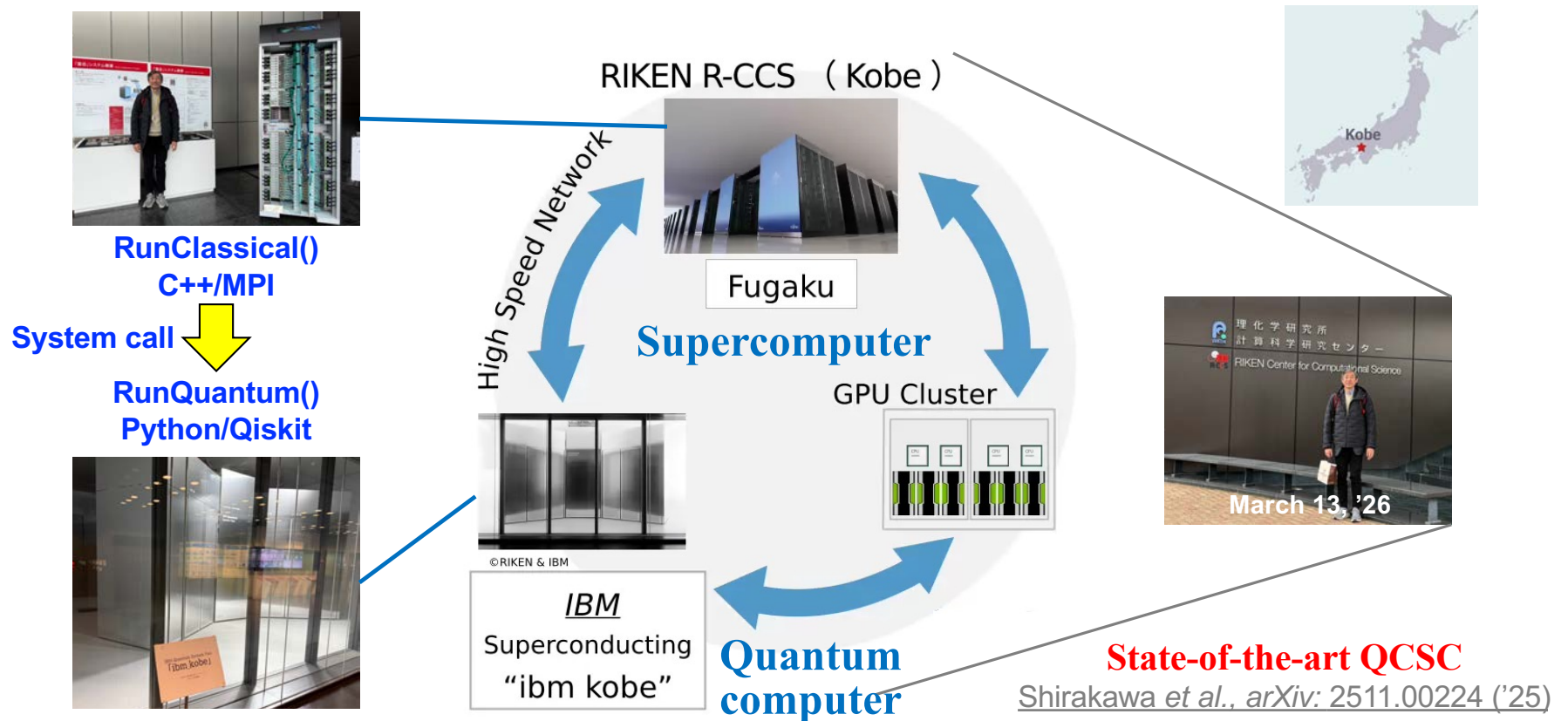


MapReduce

- Parallel programming model for data-intensive applications on large clusters
 - > User implements **Map()** and **Reduce()**
- Parallel computing framework
 - > Libraries take care of everything else
 - Parallelization
 - Fault tolerance
 - Data distribution
 - Load balancing
- Developed at Google



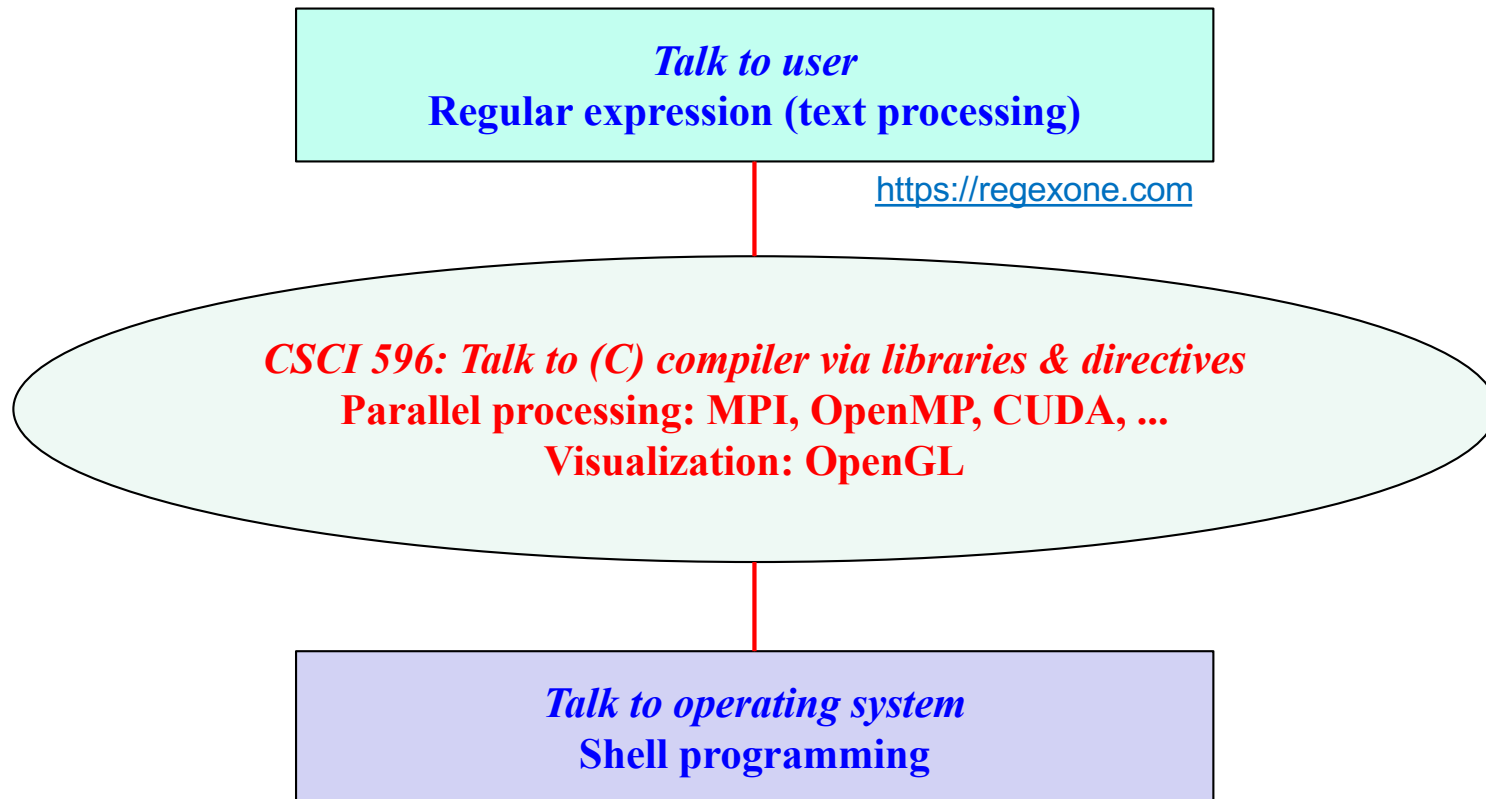
New: Quantum-Centric Supercomputing



- Many next-generation supercomputers will be tightly integrated with quantum computers as quantum processing unit (QPU), e.g., FugakuNEXT at RIKEN, Japan & NERSC at Berkeley, U.S. in ~2030

What to solve with QCSC?

Computational Research Survival Guide



A. Scopatz and K. D. Huff, *Effective Computation in Physics* (O'Reilly, '15)
<http://www.amazon.com/Effective-Computation-Physics-Anthony-Scopatz/dp/1491901535>